CIVIC VERIFICATION & INFRASTRUCTURE ACCOUNTABILITY

PARAKH

Complete User Training & Operations Manual

Citizen Reporting, Civic Verification & Infrastructure Accountability Workflow

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OWNER	Hassan (Founder & Project Lead)
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AUDIENCE	Citizens · Students · Street Representatives · Moderators · Administrators · Union Council officers · Contractors · Mayor & Operations staff · Trainers · Developers · Institutional partners

READ THIS BEFORE ANYTHING ELSE

Parakh is currently under design and development. It is **not** an official Government of Punjab application, and it must not be presented as one. No formal partnership with any government department or NGO has been signed. The complete application is **not yet fully deployed**, and **no social impact has been demonstrated**.

This manual describes **two different things** and always labels which is which:

PHASE 1 — capability designed for the supervised pilot. FUTURE — the intended institutional operating model, **not yet built and not yet approved for deployment**.

Three of the five portals in this manual — Union Council Officer, Contractor, and Mayor & Operations Command — are **future institutional capabilities**. They are documented here so that everyone understands the destination, not because they exist today.

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1. Document Control

1.1 Version history

Version	Date	Change
1.0	7 Aug 2026	First release. Complete 0–100 training and operations manual for all five Parakh portals, derived from Master Document v1.2.1. Includes the owner-directed accuracy corrections of the same date: launch language scope restated as Urdu and English with Urdu-English code-switching (Punjabi reclassified as a future extension, C-4); the unsupported adult-literacy percentage removed; every operational timeframe re-verified against the Master Document and Decision Register and labelled as a proposed pilot benchmark where it is not a finalised policy (§4.3); temporary-restriction and permanent-suspension authorities stated explicitly; and the defect liability period qualified as an owner-set 12-month default pending legal review (§8.9).

1.2 Purpose

This manual exists so that a first-time user in any Parakh role can operate their portal correctly **without verbal assistance**. It is a working operational tutorial, not a product overview, pitch deck, or requirements document.

After reading the relevant sections, a user should be able to answer, without guessing:

- What do I do, in what order?
- What information must I supply?
- What does this status mean?
- Who receives the case after me?
- What am I allowed to decide, and what am I not?
- Can the AI decide this for me? *(No.)*
- What do I do when something goes wrong?

1.3 Scope

In scope: the five operational portals; the report and issue lifecycles; evidence rules; identity and privacy rules; AI boundaries; validation, appeals and escalation; contractor evidence and defect liability; the Commit–Verify–Release model; the private operational heatmap; exception handling; role quick-starts; training scenarios; glossary.

Out of scope: software installation, server administration, source code, database schemas, procurement law, and any statement about formal government adoption.

1.4 Intended audience by section

Reader	Read at minimum
Citizen / Student	§2, §3, §5, §12 (Citizen quick start), §14 selected exceptions
Street / Community Representative	§2, §3, §6, §9 (case simulation), §12, §13
Moderator	§2, §3, §5, §6, §9, §10, §11, §14 — the whole manual, in practice
Platform Administrator	Entire manual, with emphasis on §11 privacy and §14 exceptions
Union Council Officer *(future)*	§2, §7, §9, §10, §12
Contractor *(future)*	§2, §8 (esp. §8.7 Before/After), §10, §12

Reader	Read at minimum
Mayor / Operations *(future)*	§2, §9, §10, §12
Trainer	Entire manual, plus §13 scenarios and readiness checklists
Developer	Entire manual as behavioural specification, alongside the Master Document

1.5 Related Parakh documents

Document	Role
Parakh Master Document v1.2.1	Authoritative source of truth. Where this manual and the Master Document disagree, **the Master Document governs**.
Parakh Decision Register (DR-1 … DR-18)	Approved design decisions referenced throughout as *DR-n*.
Parakh Developer & Director Handoff v1.2.1	Condensed presentation for engineering and leadership.
Open Items Register (§26 of the Master Document)	15 open dependencies, 11 of which block pilot launch.

1.6 Conventions used in this manual

PHASE 1 Designed for the supervised pilot.

FUTURE Intended institutional model. Not built, not deployed.

[UI LABEL TO BE CONFIRMED] The exact on-screen wording is not yet fixed.

[WORKFLOW DETAIL TO BE CONFIRMED] Behaviour not yet specified in the Master Document.

[POLICY DECISION REQUIRED] Awaiting an owner or institutional decision.

[NOT YET IMPLEMENTED] Described for completeness; no working implementation.

> **Honesty rule applied throughout.** Where the Master Document does not specify a button name, screen layout, threshold, or rule, this manual says so rather than inventing one. A placeholder is not a gap in the training — it is a decision that has not yet been made, and pretending otherwise would train people incorrectly.

Screen placeholders. No final user interface exists yet. Screens are therefore referenced as numbered conceptual placeholders — for example **[SCREEN PLACEHOLDER C-05 — Live Capture]** — followed by a specification of what that screen must let the user understand or accomplish.

2. Parakh in Principle

2.1 How to use this manual

Read §2 and §3 first, whatever your role — they contain the rules that apply to everyone. Then read your own portal section in full. Then read §9, the complete end-to-end case simulation, which shows how your work connects to everyone else's. Trainers should add §13.

If you only have ten minutes, read your role's quick-start guide in §12 — but do not treat that as sufficient for operational access.

2.2 What Parakh is

Parakh (Urdu: **پرکھ** — to test, to appraise, to prove worth) is a civic verification and infrastructure accountability platform.

A citizen photographs a public problem, describes it aloud in Urdu or English — or a mix of the two — and submits it. The system turns that into **sealed, verifiable evidence**. AI helps read and sort it. **Trained people decide what it means.** A representative confirms it on the ground. The responsible authority acts. The citizen is told the outcome and confirms or disputes it. Every consequential step is recorded in a history that can be checked afterwards.

The core distinction:

> Most civic platforms digitize the **complaint**.
> Parakh digitizes the **proof**.

A complaint is an assertion — it can be dismissed, forgotten, or moved between offices. Structured evidence, linked to a time, a place, supporting media and an accountable human review, is much harder to ignore.

2.3 The operational chain

Everything in this manual is one of these seven links:

> **EVIDENCE → VERIFICATION → STRUCTURED INCIDENT → RESPONSIBLE ACTION → RESOLUTION EVIDENCE → HUMAN CONFIRMATION → TRACEABLE CLOSURE**

Evidence is machine-assisted, human-confirmed, privacy-conscious and auditable at every link.

2.4 What Parakh offers, in plain terms

Evidence-first reporting	The unit of record is sealed evidence captured at the scene — not a free-text complaint typed later.
Voice-first accessibility	A person who cannot type, or cannot type Urdu, is a first-class user. The core reporting path is completable by voice and camera alone. **Launch language scope: Urdu and English, including Urdu-English code-switching.** Punjabi and other regional languages are planned under a multilingual-extensible architecture and are not a Phase-1 commitment (C-4).
Human verification	Nothing becomes "verified" because software was confident. An independent person confirms it.

Traceability	Each case carries a history that survives staff turnover, denial, and administrative handoffs.
Privacy	Reporting is verified but protected. Identity is separated from the case record and is not shown to downstream users.
Accountability	Completion is checked against evidence, not paperwork. Recurrence can be linked back to earlier work.
Infrastructure intelligence	Aggregated, privacy-protected patterns show where problems cluster and recur.

2.5 What Parakh is NOT

Not an emergency service. Parakh is not for medical emergencies, fire, crime in progress, or immediate danger to life. In an emergency, contact the relevant emergency service directly. Parakh is for civic infrastructure problems that need to be recorded, verified and resolved — not for incidents needing an immediate response.

Not an anonymous accusation platform. Reporting requires a verified identity (see §11). Identity is *protected*, not *absent*. Parakh is not a channel for untraceable allegations against named individuals.

Not an AI decision-maker. AI transcribes, suggests and compares. It never accepts, rejects, merges, validates, prioritises, approves, or closes anything. No confidence score changes that.

Not a replacement for procurement systems. Parakh does not run tenders, hold contracts, or replace any government procurement process.

Not responsible for selecting contractors. The relevant government or procurement authority selects contractors under its own rules. Parakh may *record* an authorised assignment where institutional integration exists.

Not a bank, treasury or payment custodian. Parakh never holds, pools, stores or independently transfers public money. It has no account and no balance.

Not a system that releases government funds. In the future model, an authorised officer records a decision inside Parakh; the payer's own external finance system executes any transfer (see §10).

Not an official Government of Punjab application — unless and until it is formally adopted, which has not happened.

Not a public surveillance system. There is no public live map during the pilot. The operational heatmap is private, role-restricted, aggregated and geo-fuzzed. It exists to show where problems cluster, not to watch people.

2.6 Current status, stated plainly

Item	Status
Design baseline	**Final** — 18 of 18 Decision Records approved
Application	**Not yet fully built**
Pilot	**Not approved to launch** — 11 launch-blocking open items remain
Government / NGO partnership	**None formalised**
Demonstrated impact	**None** — the pilot exists to find out whether the approach works

Anyone presenting Parakh externally must repeat these five facts accurately. The research design deliberately allows a negative result to be published.

3. The Five-Portal System

3.1 The complete ecosystem

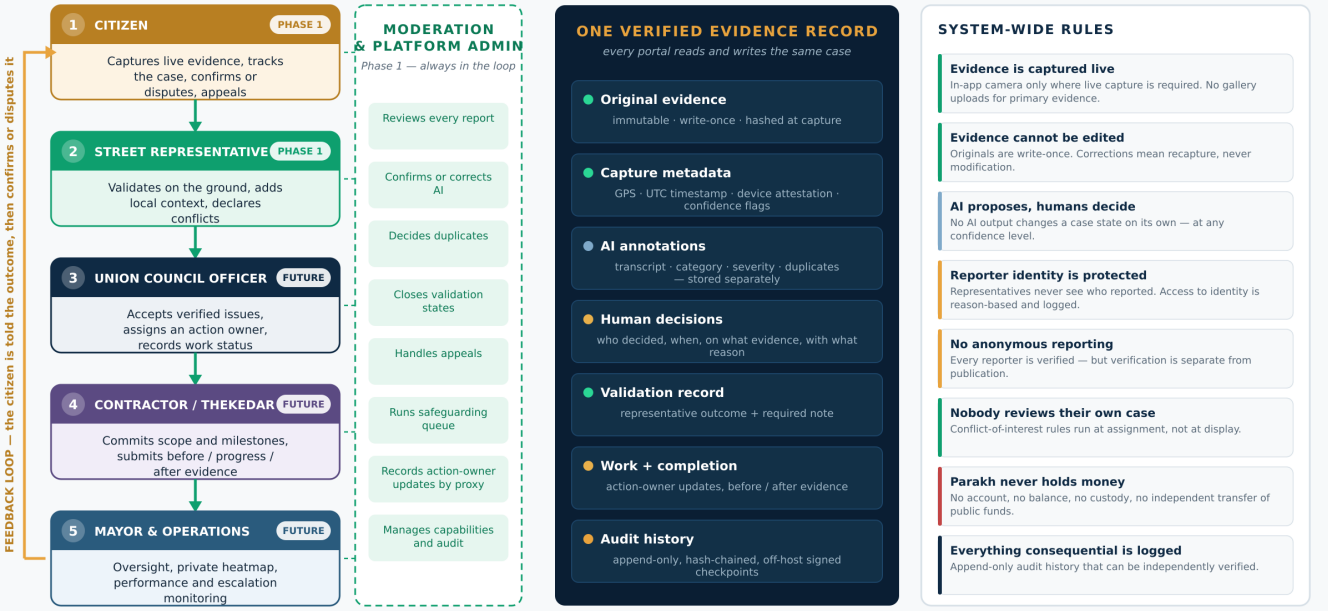


Figure 3.1 — The five Parakh portals, the moderation spine that is present at every step in Phase 1, the single shared evidence record, and the citizen feedback loop. Portals 3, 4 and 5 are future institutional capabilities.

3.2 How one issue travels

CITIZEN → STREET REPRESENTATIVE → UNION COUNCIL OFFICER → CONTRACTOR / THEKEDAR → MAYOR & OPERATIONS COMMAND
…and then **back to the CITIZEN** for confirmation.

The loop only closes when the person who reported the problem is told what happened and is given the chance to confirm or dispute it. A case that is marked resolved without that step is recorded as **RESOLVED_UNCONFIRMED**, not as a success.

3.3 The master training example

One case runs through this entire manual:

A citizen discovers a dangerous pothole on a local street after rainfall.

The road surface has broken open near a school approach road. Water has collected in it. A motorcycle rider swerved to avoid it. The citizen decides to report it.

This same case is used from first report through verification, assignment, repair, before/after evidence, resolution, oversight and closure in §9. Where a different issue type behaves differently, that difference is noted — but the pothole case is the spine.

How the same workflow adapts to other categories:

Category	What changes	What stays the same
Drainage / sewerage	Severity often depends on standing water and health risk; validation may need a return visit after rain	Live capture, GPS, voice description, validation, before/after evidence
Garbage accumulation	Recurs on a collection cycle, so duplicate and recurrence handling matter more	Everything else
Damaged streets / roads	The pothole case *is* this category	—
Streetlights	Best evidenced after dark; a daytime photo may be insufficient	Live capture rules, validation, confirmation
Water supply	The visible evidence may be indirect (a dry tap, a leak) so the voice description carries more weight	Human verification, appeal rights

Whatever the category, the chain in §2.3 does not change.

3.4 Role and permission matrix

This is the authoritative training summary of who may do what. It is derived from the Master Document permission matrix and the DR-1 capability model.

Capability	Citizen / Student	Street Rep	Moderator	Platform Admin	UC Officer F	Contractor F	Mayor / Ops F
View own reports and status	✔	✔	✔	✔	✔	✔	✔
Submit a new report	✔	✔ *(as a citizen)*	✖	✖	✖	✖	✖
Capture live evidence	✔	✔ *(supplementary)*	✖	✖	✖	✔ *(work evidence)*	✖
See nearby verified issues	✔ *(de-identified, geo-fuzzed)*	✔ *(in scope)*	✔	✔	✔	Own projects only	✔
See reporter identity	Own only	**✖ Never**	Reason-based, logged	Break-glass, dual-logged	✖	✖	✖
Validate an issue on the ground	✖	✔ *(in scope, no conflict)*	✖	✖	✖	✖	✖
Correct AI category / severity	Suggest at submission	Provide context	✔	✔	✖	✖	✖
Decide duplicates / merge	Choose to support	✖	✔	✔	✖	✖	✖

Capability	Citizen / Student	Street Rep	Moderator	Platform Admin	UC Officer F	Contractor F	Mayor / Ops F
Accept / reject a report	✖	✖	✔	✔	✖	✖	✖
Request more information	Respond	✔ *(where allowed)*	✔	✔	✔	✖	✖
Confirm or dispute resolution	✔ *(reporter)*	✔ *(consulted)*	Records outcome	✔	✖	✖	✖
File an appeal	✔	✔ *(as a citizen)*	✖	✖	✖	✖	✖
Decide an appeal	✖	✖	✔ *(only if uninvolved)*	✔ *(fallback)*	✖	✖	✖
Escalate	Via appeal	✔ to Moderator	✔ to Admin	✔	✔	✔ *(delay/ variation)*	✔
Accept an issue into operations	✖	✖	By proxy, authorised	✔	✔	✖	✖
Assign an action owner	✖	✖	✖	✔ *(configuration)*	✔	✖	✖
Select the contractor	✖	✖	✖	✖	**Outside Parakh**	✖	✖
Submit completion evidence	✖	✖	By proxy, authorised	✖	✔	✔	✖
Record a payment decision	✖	✖	✖	✖	✔ F	✖	✖
Move money	**✖ Nobody in Parakh — ever**						
See contractor / project detail	✖	Limited	✔	✔	✔	Own only	✔
View private operational heatmap	✖	✖	✔	✔	✔	✖	✔
Alter original evidence	**✖ Nobody — ever**						
Alter audit history	**✖ Nobody — ever**						

Two rows deserve emphasis. Nobody — including the Platform Administrator — can alter original evidence or rewrite audit history. That is not a permission that has been withheld; it is a capability that does not exist in the system.

4. Case Status System

Parakh keeps **two separate state machines**. This is deliberate and must not be collapsed into one.

- A **Report** is one citizen's submission — one person's evidence bundle.
- An **Issue** is the canonical record of one real-world problem. It may gather several reports.

Three people reporting the same pothole produce **three reports** and **one issue**. Each reporter keeps their own report record, their own timeline and their own notifications.

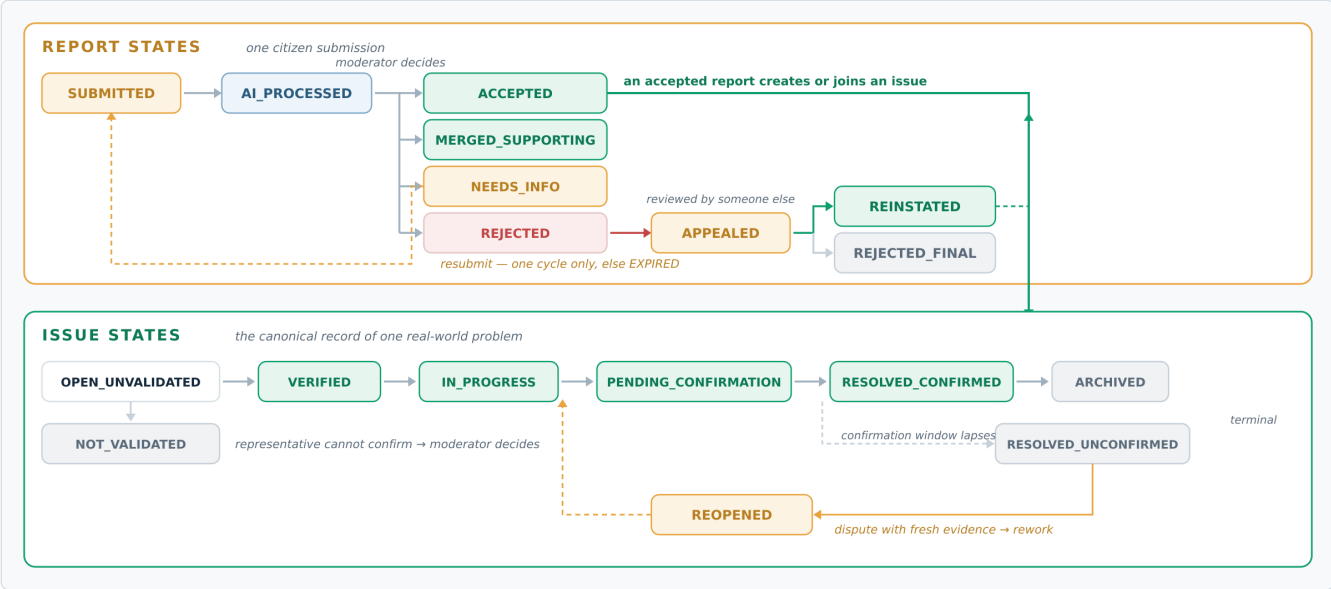


Figure 4.1 — Report states (upper lane) and issue states (lower lane), with the appeal chain and the reopen path.

4.1 Report states

State	Meaning	Who causes the transition	Evidence required	Next possible states	Notification
SUBMITTED	Evidence bundle received; capture-integrity checks passed	Citizen (T1)	Live photo(s), location, voice or text	AI_PROCESSED	Citizen: receipt + tracking ID
AI_PROCESSED	Transcript, category, severity and duplicate candidates attached as annotations	AI, assistive only (T2)	—	ACCEPTED · MERGED_SUPPORTING · NEEDS_INFO · REJECTED	Internal only
ACCEPTED	A human moderator accepted the report	Moderator (T3)	Reviewed evidence	Creates or joins an issue (T4)	Citizen: accepted
MERGED_SUPPORTING	Report describes an issue already recorded; attached as supporting evidence	Moderator (T3)	Similarity evidence shown to moderator	Follows the issue	Citizen + existing supporters

State	Meaning	Who causes the transition	Evidence required	Next possible states	Notification
NEEDS_INFO	Something specific is missing	Moderator (T3)	Guided prompt to the citizen	SUBMITTED *(one resubmission)* · EXPIRED	Citizen: what is needed
EXPIRED	The citizen did not respond within the window	System	—	Terminal *(may re-report)*	Citizen
REJECTED	Not accepted, with coded reasons	Moderator (T3)	Reason codes required	APPEALED	Citizen: reason + appeal path
APPEALED	The citizen contested a rejection or a material misclassification	Citizen files (T12)	One clarifying statement, optional extra evidence	REINSTATED · REJECTED_FINAL	Citizen: appeal received
REINSTATED	An uninvolved moderator overturned the rejection	Moderator, uninvolved	—	Re-enters the pipeline	Citizen
REJECTED_FINAL	The appeal was not upheld	Moderator, uninvolved	Final reason codes	Terminal	Citizen: final outcome

4.2 Issue states

State	Meaning	Who causes the transition	Evidence required	Next possible states	Notification
OPEN_UNVALIDATED	An accepted report has created or joined an issue; nobody has confirmed it on the ground yet	System (T4)	—	VERIFIED · NOT_VALIDATED	Scoped representatives: validation request
NOT_VALIDATED	The representative could not confirm the problem	Representative recommends, Moderator closes (T6)	Required note	May return to OPEN_UNVALIDATED with new evidence, or close *(appealable)*	Citizen: status + appeal path

State	Meaning	Who causes the transition	Evidence required	Next possible states	Notification
VERIFIED	An independent person confirmed a real, observable problem	Representative recommends, Moderator closes (T5)	Validation note; optional supplementary photos	IN_PROGRESS	Citizen, supporters, representative, action owner
IN_PROGRESS	The responsible action owner has acknowledged or started work	Action owner (T7)	Acknowledgement or work-start record	PENDING_CONFIRMATION	Citizen, supporters, representative
PENDING_CONFIRMATION	A fix has been claimed with completion evidence	Action owner (T8)	Completion evidence attached	RESOLVED_CONFIRMED · RESOLVED_UNCONFIRMED · REOPENED	Reporter(s) + representative: confirmation request
RESOLVED_CONFIRMED	The reporter confirmed the fix	Reporter confirms, Moderator accountable (T9)	Citizen confirmation	REOPENED *(within window)* · ARCHIVED	All participants
RESOLVED_UNCONFIRMED	The confirmation window lapsed with no response and no dispute	System (T10)	—	REOPENED *(within window)* · ARCHIVED	Participants: closed-unconfirmed
REOPENED	Someone disputed the resolution with fresh evidence	Reporter or Representative (T11)	**New evidence required**	IN_PROGRESS *(rework)*	Action owner, moderator
ARCHIVED	Terminal state after the dispute window closes	System	—	—	—

4.3 Timing benchmarks

> **Read this before using any number below.** Every duration in this table is a **proposed pilot evaluation benchmark**, not a finalised policy, not a guarantee, and not a service-level agreement. The Master Document records them as *proposed* values to be adjusted from observed pilot behaviour. Parakh measures actual performance against them and reports the difference; it does not promise them to anyone.
>
> Wherever this manual later refers to "the 72-hour benchmark", "the 7-day benchmark" or "the 24-hour route", it means the value in this table, with this same qualification attached.

Step	Benchmark	Status
AI triage (T2)	≤ 5 minutes, or the report goes straight to the manual queue	Proposed pilot benchmark
Moderation decision (T3)	48 hours	Proposed pilot target, adjusted to actual moderator capacity
Community validation (T5 / T6)	72 hours	Proposed evaluation benchmark; validation time is recorded and bottlenecks identified
Needs-info response window	7 days	Proposed workflow value
Appeal filing window	7 days	Proposed workflow value
Appeal decision	72 hours	Proposed workflow value
Resolution confirmation window (T9 / T10)	7 days	Proposed workflow value
Dispute / reopen window (T11)	14 days	Governance mechanism; reviewed after pilot data
Action-owner acknowledgement (T7)	7 days	**Reported as a metric, not enforced in Phase 1**
Safeguarding escalation	24 hours	The safeguarding **route** is an approved requirement; the 24-hour window is a proposed benchmark

What is settled, and what is still a benchmark

Rule	Status in the Master Document
Safeguarding escalation route — student-related welfare content goes to a restricted queue and is escalated to an administrator and the named school liaison; Parakh staff do not investigate welfare concerns themselves	**Approved requirement.** The 24-hour window itself is a proposed benchmark
Temporary account restriction — moderators may apply it; reason recorded; takes effect immediately; appealable	**Approved**
Permanent account suspension — **administrator only.** A moderator can never permanently suspend an account	**Approved**
Offline capture with an encrypted device queue and automatic upload on reconnection, with integrity metadata sealed at capture and **capture time**, not upload time, treated as the timestamp	**Approved**

Rule	Status in the Master Document
Append-only, hash-chained audit history with off-host signed checkpoints, a verification tool, documented key custody, and discontinuity records instead of chain repair	**Approved for Phase 1.** It detects change to the audit history — it does not prove a photograph is truthful or a decision correct
Launch language scope — Urdu and English including Urdu-English code-switching; other languages extensible but not committed	**Approved**

4.4 Terminal states

EXPIRED, **REJECTED_FINAL** and **ARCHIVED** are terminal. A citizen may always submit a *new* report about a problem that still exists; a terminal state closes a record, not a right.

5. Portal 1 — Citizen Portal PHASE 1

This is the portal most people will use. It is designed so that a person who cannot type — or cannot type Urdu — can still report a problem completely.

5.1 Before you start

What you need: an Android phone [minimum Android 8.0 — to be confirmed at build], camera and microphone permission, and a verified account.

What you do not need: an internet connection at the moment of capture. You can capture a full report offline; it uploads when you have signal.

5.2 Opening Parakh and choosing a language

[SCREEN PLACEHOLDER C-01 — Welcome & Language]

This screen must let the user: choose **Urdu** or **English**; understand in one sentence what Parakh is for; and proceed to account creation or sign-in. Urdu must render right-to-left correctly. The choice must be changeable later in settings without reinstalling.

Action	Choose your language
What you see	Two clear language options [UI LABEL TO BE CONFIRMED]
What you do	Tap Urdu or English
Information required	None
What Parakh checks	Nothing — this is a preference
What AI may suggest	Nothing
What a human must confirm	Nothing
What happens next	The whole app, including notifications, uses that language
Status change	None
Notification	None
If it fails	The app falls back to the device language. You can change it in settings.

5.3 Creating an account and verified identity

Parakh does **not** accept anonymous reports. This is a deliberate design decision (Master Document constraint C-6). It exists so that evidence carries accountability and so that the pilot produces trustworthy civic data. **Verification is separate from publication** — being verified does not mean being visible.

There are **two identity tracks**.

5.3.1 Adult track

Action	Verify your identity as an adult
What you see	An identity step explaining who confirms your identity and why [UI LABEL TO BE CONFIRMED]
What you do	Complete the verification path offered by your pilot site
Information required	The approved adult verification basis is **institutional attestation** — a participating school, NGO, community partner or approved pilot authority confirms you are a known real person, and that attestation is recorded in the identity vault (DR-9)
What Parakh checks	That an attestation record exists before the account can submit reports
Important	**Phone possession alone is not identity verification.** A phone OTP may be used as an authentication and account-recovery method, but it does not by itself establish who you are, and a phone number is never treated as a permanent unique person identifier
If you have no personal phone	An **institution-issued credential path** is available — a username plus a secure temporary activation credential, followed by a secret you set yourself
What a human must confirm	The attesting institution confirms your identity
What happens next	Your account is activated and can submit reports
If it fails	You cannot submit until verification completes. Contact your pilot-site liaison.

5.3.2 Student track (participants under 18) PHASE 1

Students are protected by a stricter set of rules (constraint C-8, DR-9).

Rules that must never be broken for student accounts

- **No CNIC is ever requested from a student.**
- **No personal phone number is required from a student.**
- Verification is by **school-roster attestation** plus **guardian and/or school consent**, and the student's own **assent** — a child who does not wish to take part is not enrolled by an adult's signature alone.
- Credentials are **issued and reset only by the school liaison**, through a defined process.
- One student must never use another student's credential.
- Student identity data lives in the **isolated identity vault**; the reporting system uses **pseudonymous participant identifiers only**.
- **Roster imports are minimised.** An unrestricted full school roster must not be loaded into the operational reporting database.
- On a shared device, accounts and their encrypted offline queues are **fully isolated** from each other.

Action	Student onboarding
What you see	Age-appropriate language explaining what is collected, what happens to a report, and that no personal details appear publicly
What you do	Confirm your assent after your guardian and/or school has given consent
Information required	School-roster attestation; recorded consent artefact; student assent
What Parakh checks	That an active consent artefact exists before the account is activated
What happens next	Student mode is enabled: identity hidden on every shareable surface, school-configured quiet hours applied to notifications
Withdrawal	Consent can be withdrawn at any time. The account is deactivated within 24 hours, the device queue is erased by destroying its encryption key, and the data is excluded from future model training

5.4 Privacy and consent explained to the citizen

What you see	A plain-language consent screen in your chosen language
What it must tell you	What is collected; that AI assists but people decide; whether an external AI provider processes evidence and for what purpose; that participation is voluntary; that you may withdraw without penalty
What you do	Read and give consent
What is protected	Your name and contact details are stored separately from your reports. Street Representatives **never** see who reported. Access to your identity by a moderator or administrator must have a reason and is logged
If you refuse	You cannot submit reports, and nothing is collected

5.5 Home screen

[SCREEN PLACEHOLDER C-02 — Citizen Home]

This screen must let the user: start a new report in one tap; see their own reports with current status; open any report's timeline; see nearby verified issues (de-identified and geo-fuzzed); reach settings and help. Student mode hides all identity data on this screen.

5.6 Starting a new report

Action	Start a report
What you see	A guided flow: **capture → describe → review AI preview → submit**
What you do	Tap to begin
What Parakh checks	Session validity, camera and location permissions, connectivity (offline is fine)
What happens next	The in-app camera opens
If permissions are denied	See §14.3 and §14.4 — the app must explain why the permission is needed and offer a route to settings

5.7 Capturing live photo evidence

[SCREEN PLACEHOLDER C-03 — Live Capture]

This screen must let the user: take between one and five photographs using the in-app camera; see clearly that live capture is in use; see that location and time are being attached; retake a photo before continuing. It must **not** offer any path to select an existing image.

Action	Photograph the pothole
What you see	A live camera view inside Parakh, with an indication that this is a live capture
What you do	Take 1–5 photos showing the problem and enough surroundings to identify the place
Information required	At least one photo
What Parakh checks	That the image came from the in-app camera; it computes a content hash over the exact image bytes and a versioned capture manifest at the moment of capture
What AI may suggest	Nothing yet
What a human must confirm	Nothing yet
What happens next	The photo is written into the encrypted local queue on your device
If it fails	If the camera cannot open, see §14.3

Why old, gallery or downloaded photographs are not accepted

Parakh's entire value is that its evidence can be checked afterwards. A photograph selected from a gallery has no reliable link to a place or a time — it could be from last year, from another city, or from the internet. A live in-app capture is sealed at the moment it is taken with its location, its time and a content hash, and it cannot be edited afterwards.

There is no gallery-import path in the reporting flow. This is not a restriction that can be turned off for convenience; it is the reason the evidence is worth anything.

5.8 Time and location

Action	Location and time are attached automatically
What you see	A location indicator and a captured time
What you do	Nothing, if GPS is available
What Parakh records	GPS coordinates at the moment of photo capture; the raw device time; the last trusted server-time synchronisation; the calculated offset; how old that synchronisation is; and a resulting **timestamp-confidence status**
Why the extra time fields	A phone clock can be wrong or deliberately changed. Parakh records how much it can trust the time rather than assuming it
If the clock cannot be trusted	The record is marked `time_unverified`. This **lowers evidence confidence** and is shown to the moderator — but it **does not reject your report**. A person whose phone has been offline for a week is still a witness

5.8.1 When GPS is unavailable — assisted pin-drop

Action	Place the location manually
What you see	A map on which you can place a pin [UI LABEL TO BE CONFIRMED]
What you do	Move the pin to where the problem actually is
What Parakh records	The report is marked `location_unverified`
Consequence	Verification confidence is reduced, and **no location-sensitive decision is taken on that report without additional validation** (DR-17)
Why this exists	GPS fails indoors, in dense streets and on older devices. Blocking those reports would silently exclude the people most likely to need the platform

5.9 Recording your voice description

[SCREEN PLACEHOLDER C-04 — Voice Description]

This screen must let the user: record spoken audio up to the permitted length; see that recording is in progress; play it back; re-record before continuing; and skip to typing if they prefer. It must be usable without reading long instructions.

Action	Describe the problem in your own words
What you see	A record button and a timer
What you do	Speak naturally — in **Urdu**, in **English**, or mixing both in the same sentence. Urdu-English code-switching is expected and supported by design (C-4). Punjabi and other regional languages are a planned extension, **not a Phase-1 commitment**
Length	Up to 90 seconds [pilot benchmark — may be adjusted]
What to say	Where it is, what is wrong, how long it has been there, who it affects, and any danger it causes
What Parakh checks	That audio was recorded; the recording is sealed like any other evidence
What AI may suggest	A transcript of what you said, with a confidence value
What a human must confirm	The moderator sees both your audio and the transcript
You may also	Add an optional typed note in Urdu or English
If audio is poor	See §14.5. You can re-record before submitting, and low-confidence transcripts are flagged for a human

5.10 Reviewing the AI preview

This is the point where most citizens meet Parakh's AI. What it does here is **suggest** — nothing more.

[SCREEN PLACEHOLDER C-05 — AI Preview & Review]

This screen must let the user: read the transcript of what they said; see the suggested category and severity with a confidence indication; **change the suggested category** if it is wrong; see any "this looks like an existing issue" prompt; and continue to submit.

Action	Check what Parakh understood
What you see	Your transcript; a suggested category (for example *Roads → pothole*); a suggested severity; a confidence indication; possibly a duplicate prompt
What you do	Read it. Correct the category if it is wrong. Correct the text if the transcript misheard you
What AI may suggest	Transcript · category candidates · severity · duplicate candidates · a short evidence-grounded rationale
What AI must never do	Accept, reject, merge, prioritise, validate or close anything. No confidence level changes that
What a human must confirm	Everything consequential — a moderator reviews the report regardless of what the AI proposed
If the AI is wrong	Simply correct it. Your correction is recorded and improves the system. Being wrong at this stage costs nothing
If confidence is low	The suggestion is shown as uncertain and is **not** pre-filled as though it were settled

5.10.1 Duplicate prompt

What you see	"This looks similar to an issue already reported nearby. Do you want to support that issue instead?" [UI LABEL TO BE CONFIRMED]
What you do	Choose to support the existing issue, or continue with your own report
If you support it	Your evidence is attached to that issue, your participation is recorded, and you receive its notifications. **You are not discarded**
If you continue	Your report proceeds normally; a moderator will decide whether it is genuinely the same problem
Why this matters	Two people reporting one pothole should produce one repair, not two work orders — but both people must remain visible in the record

5.11 Submitting

Action	Submit the report
What you see	A summary of what will be sent, then a confirmation with a **tracking ID**
What Parakh checks	Content hashes are re-verified when the evidence reaches the server; a mismatch sends the item to quarantine and it is never stored as valid evidence
What happens next	Report state becomes **SUBMITTED**, then **AI_PROCESSED**
Notification	Receipt with your tracking ID
Offline	If you have no signal, the report waits, encrypted, on your device and uploads automatically later. **The capture time is preserved — not the upload time**
Important	Your device keeps the evidence until the server has definitely stored it and returned the matching hash. A momentary "success" message is not enough

5.12 Tracking your report

[SCREEN PLACEHOLDER C-06 — Report Timeline]

This screen must let the user: see the current status in plain language; see every state change with its time; see what is expected of them next; and reach the confirm, dispute or appeal actions when they apply.

What the statuses mean, in citizen language:

Status shown	What it means for you
Submitted	We have your report
Under review	A trained person is looking at it
More information needed	We need one specific thing from you
Accepted	Your report was accepted and is now part of a case
Joined to an existing case	Someone reported this too; your evidence supports the same case

Status shown	What it means for you
Being checked locally	A community representative is confirming it on the ground
Verified	An independent person confirmed the problem is real
Work in progress	The responsible party has acknowledged it or started work
Please confirm	Work is claimed complete — we need you to check
Resolved (confirmed)	You confirmed it was fixed
Resolved (unconfirmed)	The window passed without confirmation
Reopened	Someone disputed the fix with new evidence
Not accepted	The report was not accepted — you may appeal

5.13 Responding to a request for more information

What you see	A specific prompt — for example, "Please add a photo showing the whole width of the road"
What you do	Capture what is asked and resubmit
How many times	**One resubmission cycle.** If you do not respond within the window, the report expires — you can always start a fresh report
What happens next	It returns to the moderation queue

5.14 Confirming or disputing a resolution

This is the step that closes the loop, and it belongs to the citizen.

What you see	A notification that work is claimed complete, with the completion evidence
What you do	Go and look, if you can. Then confirm or dispute
To confirm	Confirm that the problem is fixed → **RESOLVED_CONFIRMED**
To dispute	You must attach a **fresh photo or voice note** — a dispute without new evidence cannot be assessed → **REOPENED**
Window	7 days to respond; 14 days to dispute after resolution
If you do nothing	The case closes as **RESOLVED_UNCONFIRMED** — recorded honestly as *not confirmed*, not as a success

5.15 Appeals

When you can appeal	If your report was **rejected**, or if it was **materially misclassified** (wrong category or severity in a way that matters)
Window	7 days from the decision notification
What you may add	One clarifying statement — voice or text — and optional additional evidence
Who decides	**A moderator who took no part in the original decision.** The system prevents the original decider from handling the appeal
Possible outcomes	Reinstated · category or severity corrected · finally rejected with written reasons
Recorded	Appeal rate and overturn rate are tracked as measures of system quality — a high overturn rate means the process needs fixing, not that citizens are troublesome

5.16 Privacy and safety instructions for citizens

- **Do not photograph people** where you can avoid it. Photograph the infrastructure problem.
- **Do not report emergencies through Parakh.** Contact the emergency service directly.
- **Do not put yourself at risk** to capture evidence. No pothole is worth standing in traffic for.
- **Your identity is protected from downstream users.** Street Representatives never see who reported.
- **Welfare concerns are handled separately.** If your report contains something suggesting a person is at risk, it is routed to a restricted safeguarding queue and escalated within 24 hours — it does not appear on ordinary screens.

5.17 Rules on false or misleading reports

Reporting is identity-backed. Deliberately false or misleading reports may lead to a temporary restriction applied by a moderator, or a permanent suspension applied by an administrator.

Two protections apply:

1. **Being wrong is not the same as lying.** An honest report that turns out to be mistaken is a normal outcome and carries no penalty.
2. **No automatic accusation.** In Phase 1 **no participant receives an automated fraud label**, and no AI score may reject, restrict or escalate a report on its own (DR-16). Any restriction is a human decision, recorded with a reason, and appealable. A **moderator may apply a temporary restriction only**; a **permanent suspension is an administrator decision**.

5.18 Technical error handling — quick reference

Problem	What to do
Camera will not open	Check permission; see §14.3
Microphone will not record	Check permission; type your description instead; see §14.4
No GPS	Use assisted pin-drop; the report is marked `location_unverified`
No internet	Capture anyway — it queues and uploads later
Upload interrupted	It resumes automatically; do not delete the app

Problem	What to do
App closed mid-report	Your captured evidence remains in the encrypted local queue
Wrong transcript	Correct the text before submitting
Wrong AI category	Change it before submitting

6. Portal 2 — Street / Community Representative PHASE 1

The Street Representative is the reason a Parakh issue can be called *verified* rather than merely *reported*. This role turns one person's claim into independently confirmed fact.

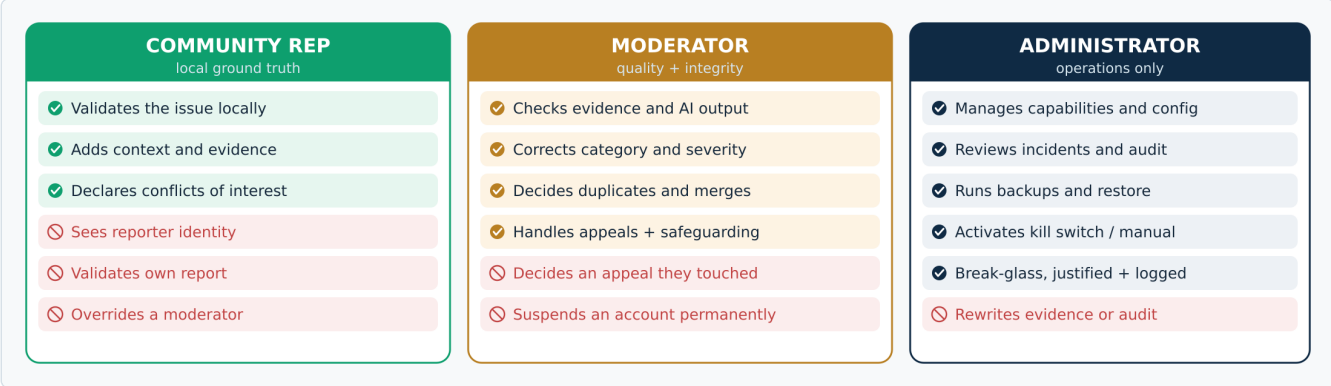


Figure 6.1 — The three Phase-1 human review roles and what each may and may not do.

6.1 Eligibility and authorisation

Representatives are **consented adults** — school staff, NGO personnel, or community members endorsed by the pilot partner. **Students are never representatives.**

Requirement	Detail
Identity	Verified through the adult track (§5.3.1)
Endorsement	By a recognised local institution, recorded in the platform
Conflict-of-interest declaration	Mandatory. Contractor relationships, political office or candidacy, financial interest in works, family relationships with platform officials. Re-attested annually
Training	Platform guidelines, evidence standards, impartial-validation practice, privacy rules, safeguarding basics
Probation	Initial validations are shadowed by a second representative or a moderator until an agreed number is reached with acceptable agreement [threshold TO BE CONFIRMED]
Grant	A **scoped, time-bounded capability grant** — a named area, with an expiry date, renewable on a fixed cycle [12 months proposed]

6.2 Secure login and scope

[SCREEN PLACEHOLDER SR-01 — Representative Sign-in]

This screen must let the user: authenticate securely; see which geographic scope they hold; and see when their grant expires.

Your **assigned geographic scope** — a school, ward or Union Council area — determines what reaches you. You will not receive validation requests from outside it, and you cannot browse cases outside it.

6.3 The validation queue

[SCREEN PLACEHOLDER SR-02 — Validation Queue]

This screen must let the user: see issues awaiting validation in their scope; see how long each has been waiting against the 72-hour benchmark; open any item; and see nothing from outside their scope.

Action	Receive a validation request
What you see	A queue of issues in **OPEN_UNVALIDATED**
What Parakh checks before it reaches you	That you hold the scope; that you did **not** report this issue; that you are not named in it; that no declared conflict of yours touches it. **These checks run at assignment, not at display** — a conflicted item never arrives
If several representatives hold the scope	Assignment is load-balanced
If you do not act	The assignment expires back to the pool after 72 hours and a moderator is notified

6.4 What you see in a validation packet — and what you never see

[SCREEN PLACEHOLDER SR-03 — Validation Packet]

This screen must let the user: view the original evidence media; read the voice transcript; read the AI summary with its confidence; see the location on a map; see related history and duplicate information; and record an outcome with a note. It must contain **no reporter identity**.

Reporter privacy — the rule that defines this role

You **never** see who reported an issue. Not their name, not their phone number, not their school roster data — not in the packet, not in the history, not in the area summary, not in a notification (DR-18).

This is enforced by removing identity **in the query that builds your packet**, not by hiding a field on screen.

Why: in a small community, knowing who reported a problem changes how it is judged, and can expose that person to pressure or retaliation. Validation is about **the problem**, not the person.

If you believe you know who reported something, that knowledge must not enter your assessment. If it affects your judgement, **recuse yourself** (§6.8).

Original evidence vs AI annotations. These are visually and structurally separate. The photographs, audio and capture metadata are the **original evidence** — immutable. The transcript, suggested category, suggested severity and duplicate candidates are **AI annotations**, stored separately and clearly attributed to a model and version. AI has never touched the original.

Confidence indicators. Where the AI's confidence is low, the packet says so. Low confidence is a signal to look harder, not a reason to reject.

6.5 Performing the validation

Action	Confirm whether the problem is real and observable
What you do	Visit the site where practical. Otherwise use documented local knowledge
What you are judging	Does this problem exist, at this place, roughly as described?
What you are not judging	Who reported it · whether it is worth fixing · whose budget it is · whether the citizen was polite
You may attach	Supplementary photographs, captured through the same live evidence pipeline as any citizen photo
You must add	A **required note** explaining your outcome. An outcome cannot be saved without one
You may add	Local context — how long it has been there, who is affected, whether it recurs, whether it is dangerous — and priority input
Outcome	**VALIDATED** or **NOT_VALIDATED**
Who closes the state	You recommend; a **moderator** closes the state (T5 / T6)
Notification	Citizen and supporters are told the issue is verified, or told it was not validated together with the appeal path

6.6 Corroborating or challenging a report

Validation is not a rubber stamp. A representative who validates everything is providing no signal at all.

Situation	What to record
The problem is exactly as described	Validate, with a note confirming what you saw
The problem exists but is **worse** than described	Validate, and add context — this may affect severity
The problem exists but is **milder** than described	Validate, and say so plainly
The problem existed but has already been fixed	Note it. [WORKFLOW DETAIL TO BE CONFIRMED — whether this closes the issue or routes to the moderator]
You cannot find the problem at that location	**NOT_VALIDATED**, with a note describing what you checked
You cannot safely reach the site	Mark it `unsafe_to_verify` — this itself raises the issue's severity for review
The location is outside your scope	Send a reroute suggestion to the moderator; do not validate it

6.7 Requesting additional information

Where permitted, you may ask for a specific missing detail rather than rejecting outright. Be precise: *"a photo showing the pothole relative to the school gate"* is useful; *"more information"* is not.

6.8 Conflict of interest and recusal

You must recuse yourself if:

- You reported the issue yourself.
- You are named in it.
- You have a declared interest that touches it — a contractor relationship, a financial interest in the works, a family relationship with an involved party.
- You discover mid-review that you know the reporter and that this affects your judgement.

How: use the recusal action. The item returns to the pool and is reassigned. **The recusal is logged.** Recusing yourself is a mark of good practice, not a failure.

You cannot silently delete or bury an inconvenient report. There is no such capability. Every action you take — and every action you decline to take — is recorded.

6.9 Suspected abuse or fraud

If you believe a report is deliberately false or malicious:

1. **Do not label the citizen.** Record what you observed and why you doubt it.
2. Route it to the **moderator** through the flag mechanism.
3. A **human** decides. **No AI model may label a citizen fraudulent** (DR-16), and no automated score may reject, restrict or escalate a report in Phase 1.
4. If a restriction results, the citizen may appeal it. A **moderator may apply a temporary restriction only**; a **permanent suspension is an administrator decision**. Either way the reason is recorded and the decision is appealable.

6.10 Escalation

Escalate when	To whom
The packet contains a welfare or safeguarding concern	**Immediately** to the moderator through the safeguarding route — 24-hour escalation applies
You have a conflict	Recuse; the system reassigns
The issue is outside your scope	Moderator, with a reroute suggestion
You disagree with a moderator's final call	Record your dissent — it is visible to administrators and feeds calibration
The site is dangerous	Mark `unsafe_to_verify`
The same problem keeps recurring	Note the recurrence; it matters for later contractor accountability

6.11 Audit trail

Every validation, note, supplementary photo, recusal and dissent is written to the append-only audit history with your identity, the time and the reason. A sample of validations is independently audited by moderators, and disagreements trigger review. Your **agreement rate** and **median validation time** are visible to you and to moderators and administrators — but they are **not published**.

6.12 Handoff to UC Operations

Once an issue is **VERIFIED**, it becomes actionable.

- In **Phase 1**, there is no statutory government actor. The action owner is the **pilot-site authority** — school administration, NGO partner or community action group — and a moderator or administrator records their acknowledgement, work-started and fix-claimed updates **by proxy**, where authorised and audited.
- In the **future institutional model**, the same verified issue reaches a **Union Council Officer** (§7) without the state machine changing at all. Only the capability assignment changes.

6.13 Representative reference card

Representative CAN

- Validate issues in their scope
- Add local context and priority input
- Attach supplementary live-captured photos
- Request additional information where allowed
- Recuse themselves
- Flag safeguarding and suspected abuse
- Record dissent from a moderator's decision

Representative CANNOT

- See reporter identity — ever
- Validate their own or a related report
- Delete or hide a report
- Overrule a moderator
- Label a citizen fraudulent
- Access the identity vault
- Alter original evidence or audit history
- Decide payments or assign contractors

ESCALATE WHEN

- Safeguarding or welfare concern → immediately
- Conflict of interest → recuse
- Outside scope → reroute suggestion
- Site unsafe → `unsafe_to_verify`
- Suspected malicious report → flag to moderator
- Persistent recurrence → note it

EVIDENCE REQUIRED

- A **required note** on every outcome
- Live-captured supplementary photos where they help
- A description of what you actually checked when recording NOT_VALIDATED

Common mistakes

Mistake	Why it matters
Validating from memory without visiting	The value of the role is independent observation
Writing "confirmed" as the whole note	The note must say what you saw
Validating a case involving a relative or business associate	This is exactly what recusal exists for
Trying to work out who reported it	Irrelevant to your judgement, and against the design
Letting items expire silently	The 72-hour benchmark exists so citizens are not left waiting
Treating high AI confidence as proof	The AI has never been to the street

7. Portal 3 — Union Council Officer / UC Operations FUTURE

> **This entire portal is a future institutional capability.** It is **not built** and **not deployed**. In Phase 1 there is no formal government authority involved, and pilot-site authority updates are recorded by proxy by an authorised moderator or administrator.
>
> The statutory mapping between issue categories and the responsible tier or department — under the **Punjab Local Government Act 2025**, the current governing framework — **must be confirmed with the partner institution and legal review** before any institutional deployment. It is an open item, not a design gap. Parakh keeps the responsible "action owner" **configurable** precisely because Punjab has enacted four local-government statutes since 2013.

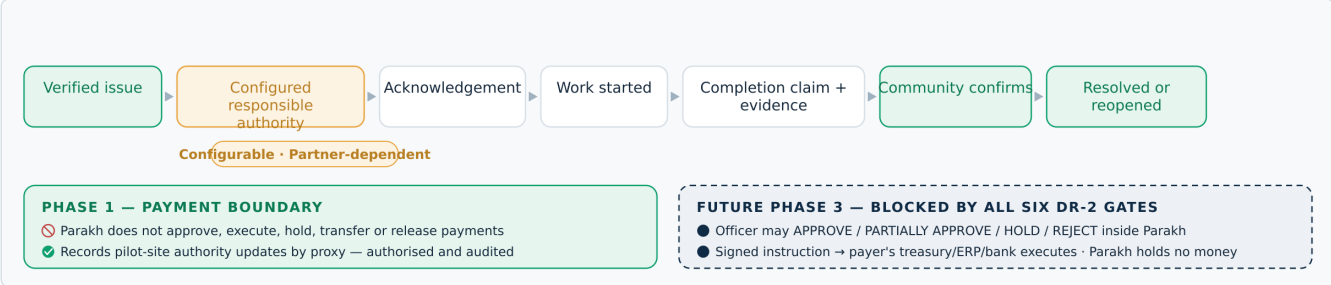


Figure 7.1 — The institutional flow, with the Phase-1 and future Phase-3 payment boundaries stated separately.

7.1 Secure login and role permissions

[SCREEN PLACEHOLDER UC-01 — Officer Sign-in]

This screen must let the user: authenticate with **multi-factor authentication** (required for privileged roles); see their jurisdiction; and see the capabilities their grant carries.

Officer capabilities are **scoped** to a jurisdiction and **time-bounded**, like every other Parakh grant. Every consequential action is audit-logged.

7.2 The UC dashboard

[SCREEN PLACEHOLDER UC-02 — UC Dashboard]

This screen must let the user: see verified issues awaiting acceptance; see issues already in progress; see items overdue against benchmarks; see anomaly flags; and reach the private operational heatmap for their jurisdiction.

7.3 The verified issue queue

Only **VERIFIED** issues reach an officer. Unvalidated reports, rejected reports and safeguarding-flagged content never appear here. That is the point of the two review layers before this one.

[SCREEN PLACEHOLDER UC-03 — Verified Issue Queue]

This screen must let the user: sort and filter verified issues by category, severity, age and location within their jurisdiction; and open the full evidence bundle for any of them.

7.4 Reviewing the evidence bundle

[SCREEN PLACEHOLDER UC-04 — Verified Issue Detail]

This screen must let the user: view all original evidence; read the transcript; see the AI annotations with confidence and provenance; read the representative's validation note; see the number of distinct verified supporters; see duplicate and cluster history; see the full audit trail; and take an operational decision.

What the officer sees	Why it matters
Original evidence	Photographs and audio, unaltered since capture
Capture metadata	Location, time, and any `location_unverified` or `time_unverified` flags
Human verification history	Who validated it, when, and what they wrote
Representative context	Local knowledge the officer would not otherwise have
Community support count	How many distinct verified people are affected
Duplicate / cluster history	Whether this is a recurring site
Audit trail	Every action taken on this case so far
What the officer does not see	**Reporter identity** — protected here as everywhere else

7.5 Accepting an issue into the operations workflow

Action	Accept a verified issue as an actionable work item
What you do	Review the bundle, then accept it into the operations workflow
What you record	The determination that this is a municipal responsibility, and which unit owns it
Status change	The issue moves toward **IN_PROGRESS** when acknowledgement or work start is logged (T7)
Notification	Citizen, supporters and representative are told
Audit	Your identity, the time and your reasoning are recorded

7.6 Assigning an action owner

The **action owner** is the party responsible for doing something about the issue. Parakh keeps this **configurable** rather than hard-coded to any office.

What you do	Assign the responsible unit or officer
What Parakh records	Who assigned it, to whom, when, and why
If the assignment is wrong	Reassign it. The correction — not just the final state — stays in the audit trail
Configurable	[Department mapping is partner-dependent and must be confirmed]

7.7 Field inspection FUTURE

Where an **Inspector / Field Verifier** role is activated in institutional phases, high-impact issues and contractor completions may receive an official on-site verification. The inspector captures evidence through the **same tamper-evident pipeline citizens use** — identical integrity guarantees — and records a signed attestation.

Conflict rules apply: an inspector may not verify works executed by an entity with which they hold a declared relationship, and assignments are randomised where inspector density allows.

7.8 Determining the municipal response

The officer determines the appropriate response and records it. Where a work item results, the officer records:

Field	Note
Work scope	What is to be done
Estimated / approved cost	Referenced against the applicable rate card [rate-card source is partner-dependent]
Budget reference	The relevant allocation
Timeline	Expected start and completion
Milestones	Where the work has more than one stage
Administrative approval	Recorded according to the authority's own rules

Anomaly detection. Cost outliers and duplicate orders are flagged automatically for human review. A flag is a prompt to look, never an accusation and never an automatic rejection.

7.9 Work orders and contractor assignment

> **PARAKH DOES NOT SELECT THE CONTRACTOR.**
>
> The appropriate government or procurement authority selects the contractor according to applicable rules and its own systems. Parakh **records or links** the resulting authorised assignment where institutional integration allows it.
>
> Parakh also does not approve contracts, enforce penalties, or replace any procurement process. The authority retains **contractor selection, contract approval and legal enforcement** in full.

7.10 What each party sees after an officer acts

Party	What they see
Citizen	Status moves to *Work in progress*; they are notified; identity remains protected
Street Representative	The same status change, plus their earlier validation in the history
Contractor F	The authorised work order with scope, timeline, milestones and quality commitment
Mayor / Operations F	The case appears in jurisdiction roll-ups, ageing and heatmap views
Audit trail	Every one of these actions, with actor, time and reason

7.11 Delays, corrections and escalation

Situation	Handling
Acknowledgement is late	The 7-day acknowledgement benchmark (§4.3, proposed) is **reported as a metric**, not enforced in Phase 1
Wrong operational unit	Reassign; the original assignment stays visible in the audit trail
Work stalls	Ageing surfaces in the UC dashboard and in Operations Command roll-ups
Dispute after resolution	The issue reopens (T11) and returns to **IN_PROGRESS** for rework
Escalation	To the next authority level per the institution's own escalation policy [POLICY DECISION REQUIRED — escalation ladder is partner-specific]

8. Portal 4 — Contractor / Thekedar FUTURE

> **This entire portal is a future institutional capability.** It is **not built** and **not deployed**. It becomes relevant only when Parakh operates alongside a formal procurement and works process.
>
> **Parakh does not select contractors, approve contracts, hold money, transfer money, or enforce penalties.** It records evidence about work, and it makes that evidence available to the authority that does hold those powers.

8.1 What this portal is for

A contractor uses Parakh to do four things, and only four:

1. **See** the authorised work assigned to them, with scope, timeline and quality commitment.
2. **Accept** the assignment and record progress against it.
3. **Submit** verifiable evidence that the work was started, progressed and completed.
4. **See** the outcome of verification, and respond to a rejection, a rework instruction, or a defect claim.

Everything else — winning the work, signing the contract, being paid, being penalised — happens **outside Parakh**, in the authority's own systems.

8.2 Authorisation, onboarding and organisation profile

Requirement	Detail
Identity	Verified organisational identity, linked to the entity named in the work order
Organisation profile	Registered entity name · registration or licence reference · contact details · authorised signatories [exact fields TO BE CONFIRMED with the partner institution]
Named operators	Individual users act on behalf of the entity; each action is attributed to a **named person**, not to a faceless company
Scope	The contractor sees **only their own** work orders. No jurisdiction-wide browsing, no other contractors' records
Authentication	Multi-factor, as for all privileged roles
Grant	Scoped and time-bounded, like every Parakh capability grant

> **Onboarding does not mean selection.** Being registered in Parakh gives a contractor no claim to any work. Selection happens in the authority's procurement process (§8.11). Parakh records the result.

8.3 Secure login and the work order list

[SCREEN PLACEHOLDER CT-01 — My Work Orders]

This screen must let the user: authenticate with multi-factor authentication; see all work orders assigned to their entity; see status, deadline and milestone position for each; filter by state; and open the full detail of any one.

[SCREEN PLACEHOLDER CT-02 — Work Order Detail]

This screen must let the user: read the scope of work; see the location; see the **before** evidence; see the start date, target completion date and milestones; see the approved cost and any material or labour estimates recorded by the authority; see the quality commitment and defect liability period; accept the assignment; and start an evidence submission.

What the work order contains	Work-order reference · linked issue · location · scope of work · **start date** · **target completion date** · milestones · approved cost · budget reference · **material estimate** and **labour estimate** where the authority records them [field availability depends on institutional integration] · quality commitment · defect liability terms · verification requirements
What the contractor also sees	The **before evidence** — the original citizen photographs and the representative's validation note
What the contractor does not see	**Reporter identity** · other contractors' orders · jurisdiction-wide data · the identity vault · the internal operational heatmap
Why the before evidence is shown	Because the completion evidence must be comparable to it. A contractor who cannot see the starting condition cannot demonstrate the change

8.4 Accepting and acknowledging an assignment

Action	Accept the assigned work order
What you do	Read the scope, dates, milestones and quality commitment, then record acceptance
What you are accepting	The scope as written · the target completion date · the milestone structure · the quality standard · the **defect liability period stated in the work order (default 12 months)**
Validation	You must be the named entity on the order; the order must be in an assignable state
System response	Acceptance recorded with your named operator's identity and the time
AI role	None
Human role	A named person accepts on behalf of the entity
Notification	The assigning officer is notified F
Audit event	Acceptance recorded
Failure path	Scope disputed → raise a query rather than accepting. [WORKFLOW DETAIL TO BE CONFIRMED — whether a contractor may formally decline inside Parakh, or only through the authority]

8.5 Before-work and progress evidence

Three kinds of evidence exist on a work order. All three use the **same live-capture pipeline** as a citizen report — the integrity guarantees do not change by role.

Evidence	When	Purpose
Before evidence	Already on file — captured by the citizen and, where added, the representative	The baseline the completion evidence is measured against

Evidence	When	Purpose
Site-condition evidence	Optional, at the point work starts	Lets the contractor record the condition they actually found, which protects them if it differs from the citizen photograph
Progress evidence	At each milestone, where the work has more than one stage	Shows the work advancing, not just appearing finished

Action	Record work start, and progress at each milestone
What you do	At the site, open the work order, select the milestone, capture live photographs, optionally add a voice note
Validation	Live capture · GPS proximity · timestamp within the work window · integrity hash
System response	Evidence attached to the milestone and to the issue
Status change	The issue is already **IN_PROGRESS**; progress evidence does not change the state, it enriches the record
Why it matters	A single after-photograph on a multi-stage job proves only that something was done at the end. Milestone evidence is what makes staged verification possible
Failure path	Site condition materially differs from the before evidence → capture it and raise a variation query (§8.6) **before** proceeding

8.6 Milestone updates, delays and variation requests

Delays	Record the delay against the milestone with a **written explanation**. A recorded, explained delay is a normal operational fact; an unexplained silence is what damages a performance record
What Parakh does with it	Surfaces the delay in the officer's queue and in Operations Command ageing views F
What Parakh does not do	Apply a penalty. Penalties are contractual and belong to the authority
Variation / change requests	Where the work as found differs from the work as scoped, the contractor raises a variation request with evidence
Who decides a variation	The **authorising officer**, not Parakh and not AI. [WORKFLOW DETAIL TO BE CONFIRMED — variation approval routing depends on the institution's own delegation rules]
Audit event	Every delay, explanation and variation request is recorded with actor, time and reason

8.7 Submitting completion evidence — the Before/After check

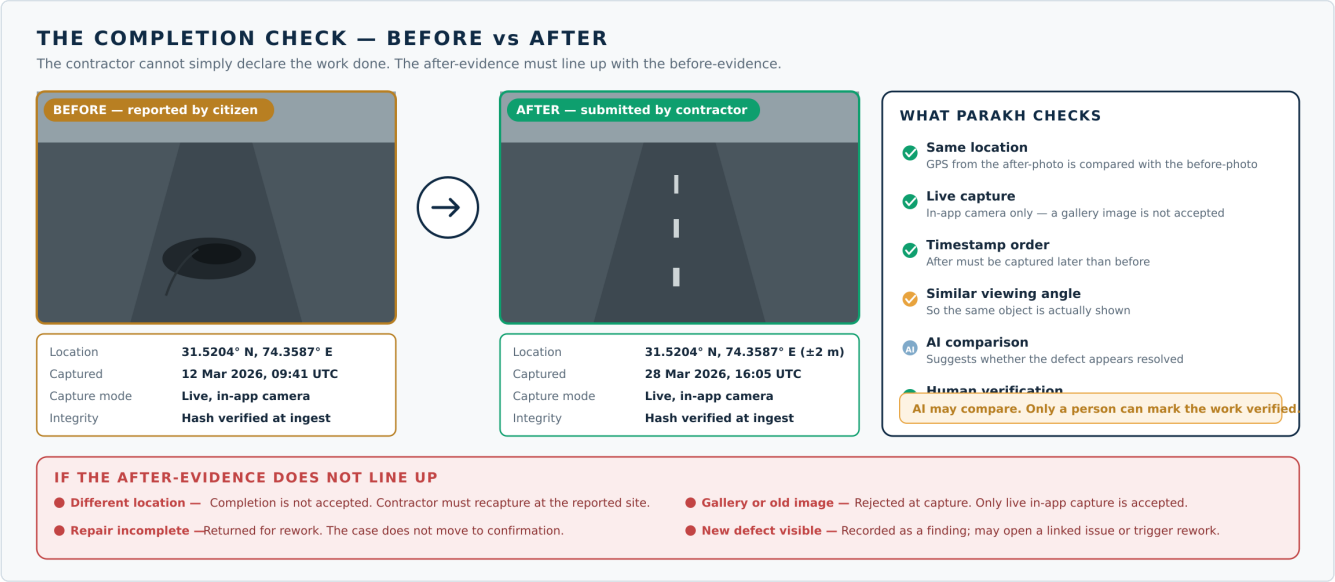


Figure 8.1 — What Parakh checks when completion evidence is compared against the original evidence, and what happens when a check fails.

The completion evidence rule — non-negotiable

Completion evidence must be **captured live inside the application**, from the **same location** and, where the original allows it, from the **same or a sufficiently similar viewing angle** as the before evidence.

Uploading a photograph from the gallery is not permitted. There is no gallery upload path.

An after-photo taken from a different street, a different angle, or a different day tells you nothing about whether *this* problem was fixed. The comparison **is** the verification.

Action	Submit completion evidence for a work order or milestone
What you do	Go to the site. Open the work order. Start evidence capture. The app shows the before-photo as a **reference overlay or side-by-side guide**. Position yourself to match it. Capture
Input	Live photographs (and video where the work type requires it) · optional voice note describing what was done · milestone selection
What Parakh checks automatically	Live capture (not gallery) · **GPS proximity** to the recorded issue location · **capture timestamp** within the work window · **viewing-angle similarity** to the before evidence · evidence **integrity hash** · metadata completeness
If a check fails	The submission is **flagged for human review** — it is not silently accepted, and it is not silently destroyed. Location mismatch, timestamp anomaly, angle mismatch and integrity failure each carry their own flag
AI role	Compare before and after; **describe** observable differences; flag apparent mismatches; estimate visual completeness. **Advisory only**
Human role	A **human decides** whether the work is acceptable. AI does not approve work, does not reject work, and does not certify completion
Status change	Issue moves to **FIX_CLAIMED**
Notification	Citizen, supporters and representative are notified that a fix has been claimed and are invited to confirm
Audit	Submission recorded with the submitting person's identity, time, device attestation and all check results

A claimed fix is not a confirmed fix. FIX_CLAIMED is a claim. Only human confirmation moves an issue to **RESOLVED**. This distinction is the entire point of the platform.

8.8 When completion evidence is not accepted — rework

What was found	What happens
After image from the wrong location	GPS check fails → flagged → human review → resubmission required from the correct location
After image from a materially different angle	Similarity check flags it → human review → resubmission where the original angle is reproducible
After image outside the work window	Timestamp flag → human review; the contractor explains
Repair visibly incomplete	Human reviewer records **incomplete**; the issue stays **IN_PROGRESS**; rework instruction issued
A different defect is visible in the after image	Recorded as a **new observation**. It does not silently become part of this work order — a new issue or a variation is raised, with a named human deciding which
Citizen disputes the claimed fix	Issue returns to **IN_PROGRESS** (T11). Rework, not argument
Evidence integrity hash fails	Treated as tampering evidence: flagged, retained, escalated for human review

Rework loop	The contractor returns to the site, completes the outstanding work, and submits **fresh live evidence**. The earlier submission is **never deleted** — it stays in the record alongside the reason it was not accepted
Why the failed submission is kept	A performance record built only from successful submissions is not a record. The rejections are the signal
Right of reply	The contractor may attach a written response to any rejection

8.9 Quality commitment and defect liability

A contractor executing work under this model accepts, as part of the work order:

Commitment	Meaning
Standards compliance	Work executed to the specification and standards named in the order
Defect liability period (DLP) — default 12 months	Any failure caused by poor execution during the liability period is **corrected at the contractor's own cost**. The 12-month figure is an **owner-set default, pending legal review and confirmation in the applicable work order or contract**; the contracting authority may specify a different period
Evidence obligation	Verifiable before, progress and after evidence, captured live
Traceability	Every submission attributable to a named person

Action	A defect is reported during the liability period
Trigger	A citizen reports the same problem at the same location; Parakh links it to the closed case through location and category clustering
What happens	The linked defect claim is raised against the work order and the responsible authority is notified
AI role	Detect and propose the link. **Advisory**
Human role	A human confirms whether it is the same failure, a new problem, or ordinary wear
What Parakh does	Records the finding and makes it available to the authority
What Parakh does not do	**Enforce** the liability. Enforcement is contractual and belongs to the authority. Parakh supplies the evidence; the authority applies the consequence

8.10 DLP monitoring, recurrence and audit history

[SCREEN PLACEHOLDER CT-03 — Work History & Liability]

This screen must let the user: see completed work orders still inside their defect liability period; see any defect claims raised against them; see the evidence behind each claim; attach a documented response; and see the full audit history of their own submissions.

Recorded	Why
DLP end date per completed work order	So both sides know when the obligation lapses

Recorded	Why
Defect claims raised, with linked evidence	So a claim can be examined, not just asserted
Recurrence at the same location	Distinguishes a genuine execution failure from a new, unrelated problem
Rework submissions	Shows whether a claim was answered
Full audit history of the contractor's own actions	So the contractor can see exactly what is on their record

Recurrence is not automatically the contractor's fault. A road can fail again for reasons that have nothing to do with the repair — a burst main beneath it, an unusual flood, heavy vehicles it was never specified for. **A human decides** whether a recurrence is a defect. No model, no score, and no automatic rule assigns blame.

8.11 Who selects the contractor — and who does not

PARAKH DOES NOT SELECT THE CONTRACTOR.

The appropriate government or procurement authority selects the contractor according to applicable rules and its own systems. Parakh **records or links** the resulting authorised assignment where institutional integration allows it.

Parakh also does not approve contracts, enforce penalties, or replace any procurement process. The authority retains **contractor selection, contract approval and legal enforcement** in full.

8.12 Contractor performance record

Parakh maintains a factual performance record from verified events only:

Recorded	Not recorded
Work orders completed and verified	Subjective opinion
Time to completion against commitment	Unverified allegations
Recorded delays and their explanations	Political commentary
Evidence rejection count and reasons	Any AI-generated character judgement
Defect claims raised and their resolution	Any automated fraud score
Recurrence at previously "fixed" sites	

Visibility.

[POLICY DECISION REQUIRED — whether contractor performance records are public, authority-only, or tiered. This has real legal and reputational consequences and must be decided w

Right of reply. Any contractor whose record shows a negative finding must be able to attach a documented response. A record that cannot be answered is not a fair record.

8.13 Contractor reference card

Contractor CAN

- See their own assigned work orders
- See the before evidence for those orders
- Accept and acknowledge an assignment
- Record site condition and progress evidence
- Record a delay with a written explanation
- Raise a variation request with evidence
- Submit live-captured completion evidence
- Respond to a rejection or a defect claim

Contractor CANNOT

- See reporter identity
- Upload photographs from a gallery
- See other contractors' work
- Confirm their own work as resolved
- Edit or delete submitted evidence
- Alter the before evidence
- Change status, cost or timeline records
- Approve their own variation
- Receive money through Parakh

EVIDENCE REQUIRED

- Live capture, in-app, no exceptions
- Same location as the original
- Same or sufficiently similar angle
- Within the work window
- Per milestone where milestones exist
- Integrity hash intact
- A written explanation for every delay

COMMON MISTAKES

- Photographing from a convenient angle instead of the matching one
- Submitting after leaving the site
- Submitting one photo for multi-stage work
- Letting a delay pass unexplained
- Treating FIX_CLAIMED as completion
- Assuming a passed automated check means approval

Escalation path. Scope query or variation → assigning officer. Disputed rejection → the officer's review, then the authority's appeal route. Suspected error in a defect claim → attach a documented response and escalate to the authority. **Security reminder:** sign out on shared or site devices; every action carries your name.

9. The Complete Case Walkthrough — 0 to 100 PHASE 1 + FUTURE

This section follows **one pothole** from the moment a citizen sees it to the moment the case is closed and archived — every screen, every field, every check, every notification, every audit entry, and what happens when each step fails.

The case

After two days of monsoon rain, a pothole has opened on the street outside a school in Ladheywala Virkan, Punjab. It is roughly half a metre across and deep enough that a motorcycle wheel drops into it. Children walk past it twice a day. Two people have already fallen.

Aisha, 34, a shopkeeper whose children attend the school, decides to report it.

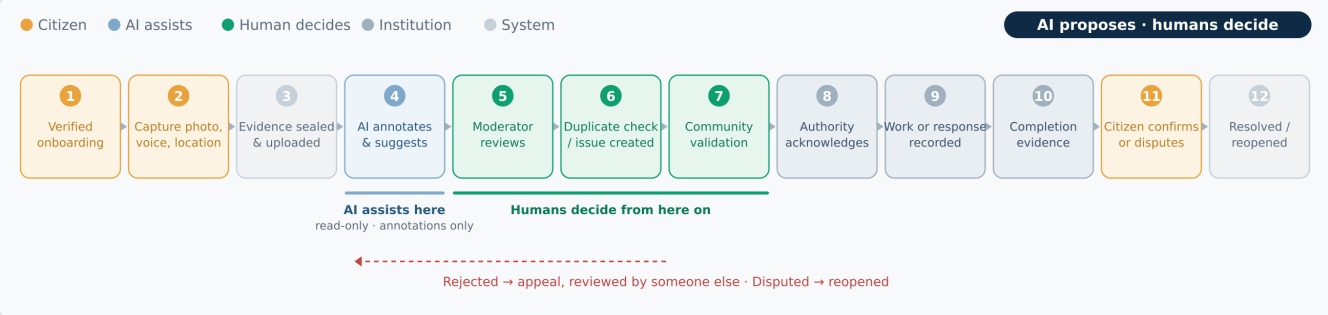


Figure 9.1 — The full lifecycle this walkthrough traverses.

How to read this section. Names, times and reference numbers are **illustrative**. The **steps, states, checks, roles and rules** are taken from the approved design. Where a step depends on something not yet decided, it is tagged rather than invented.

Phase A — Reporting

STEP 001 — Opening Parakh

Role	Citizen (Aisha)
Screen	Home [C-01]
Action	Open the application
Input	None
Validation	Session valid · account active · not restricted
System response	Home screen with the primary report action and Aisha's own case list
AI role	None
Human role	Decides to report
Status before	—
Status after	—

Notification	None
Audit event	Session opened
Next responsible role	Citizen
Failure path	Not signed in → sign-in. No network → the app still opens; capture works offline (§9, STEP 011)

STEP 002 — Identity check

Role	Citizen
Screen	Identity status [C-02]
Action	Confirm the account is verified
Validation	**Anonymous reporting is not permitted (C-6).** Every report carries a verified identity in the vault
System response	Verified → proceed. Unverified → verification flow first
Human role	Completes verification if needed
Audit event	Identity state read
Failure path	Verification pending → the report can be drafted but not submitted [WORKFLOW DETAIL TO BE CONFIRMED]

> **Why no anonymous reports.** A verified identity is what makes an evidence record credible enough to act on, and it is what makes the platform resistant to coordinated abuse. The identity is held in a **separate encrypted vault** and is **never shown to reviewers, officers, contractors or the public** (DR-9, DR-18). Verification and exposure are different things.

STEP 003 — Starting a report

Role	Citizen
Screen	Home → Capture [C-03]
Action	Start a new report
System response	Camera opens **directly**. There is no gallery option
AI role	None
Status after	Report enters **DRAFT**
Audit event	Draft created
Failure path	Camera permission denied → the app explains why live capture is required and offers to open settings

STEP 004 — Live photo capture

Role	Citizen

Screen	Capture
Action	Photograph the pothole
Input	1–3 live photographs. Aisha takes three: close-up for depth, mid-shot showing the road, wide shot showing the school gate
Validation	**Live capture only** · device attestation · integrity hash computed at capture · GPS captured · timestamp captured
System response	Media stored **write-once**. Any later attempt to alter it breaks the hash and is detectable
AI role	None yet — AI never runs before capture and never touches the original
Human role	Frames the shot
Notification	None
Audit event	Media captured, with hash
Failure path	Attempted gallery upload → **blocked, no exception**. Attestation failure → flagged for human review, not deleted

> **Why live capture only.** A gallery upload can be any image from any place at any time. A live capture, hashed at the moment it is taken with location and time attached, is a **claim about a specific place at a specific moment**. Only the second kind of image can support an accountability decision.

STEP 005 — Location capture

Role	Citizen
Screen	Location confirm [C-04]
Action	Confirm or adjust the location
Input	GPS reading, adjustable on a map
Validation	Accuracy radius checked. Weak or absent signal → the report is marked `location_unverified` and is **still accepted**
System response	Location stored with its accuracy and its verification flag
AI role	None
Human role	Corrects the pin if it is wrong
Audit event	Location recorded, with accuracy and flag
Failure path	GPS unavailable → manual placement, flagged `location_unverified`. **A weak signal never blocks a report**

STEP 006 — Voice description

Role	Citizen
Screen	Voice note [C-05]
Action	Describe the problem by speaking
Input	Aisha speaks in Urdu — Punjabi-inflected, as most people in her village speak it — code-switching into English for *"school"* and *"motorcycle"*. Parakh handles this as **Urdu with English code-switching**, which is the launch language scope
Validation	Audio recorded and stored as evidence in its own right
System response	Audio saved; transcription is queued to run **after** submission, not before
AI role	None at this moment
Human role	Speaks naturally. **No form to fill in, no category to choose, no literacy requirement**
Audit event	Audio captured, with hash
Failure path	No microphone permission → typed description offered as a fallback. Voice is the **primary** path, not the only one

Why voice first. Literacy and digital-access barriers remain significant across Punjab, and everyday speech mixes Urdu and English within single sentences. A form that requires typing and choosing a category **excludes exactly the people most affected by broken infrastructure**. Voice-based reporting is the design's answer to that, and it is why the core path can be completed with camera and voice alone.

Launch language scope. Urdu and English, including Urdu-English code-switching, at launch. Punjabi and other regional languages are planned under Parakh's multilingual-extensible architecture and are **not a Phase-1 commitment** (C-4). A speaker whose Urdu is Punjabi-inflected is a fully supported user; a report spoken entirely in Punjabi is not something the launch system claims to transcribe.

STEP 007 — Submitting

Role	Citizen
Screen	Review and submit [C-06]
Action	Submit the report
Validation	At least one live photo · location present (verified or flagged) · description present · identity verified
System response	Report accepted; a **case reference** is issued; the evidence bundle is sealed
Status before	**DRAFT**
Status after	**SUBMITTED** (T1)
Notification	Confirmation to Aisha, with her case reference
Audit event	Report submitted — actor, time, bundle hash
Next responsible role	The system (AI processing)

Failure path	Missing required element → a specific message naming what is missing. Offline → queued (STEP 011)

Phase B — AI Processing



Figure 9.2 — Everything AI does in this phase is advisory. Nothing here changes a state.

STEP 008 — Transcription

Role	AI system
Action	Transcribe the voice note
Input	The audio evidence
System response	Transcript stored **as an annotation**, separate from the original audio, tagged with model and version
AI role	Speech-to-text over **Urdu and English, including Urdu-English code-switching** — the approved launch scope (C-4)
Human role	None yet. A **reviewer always sees the transcript next to the original audio** and can rely on the audio
Status before	**SUBMITTED**
Status after	**PROCESSING** (T2)
Audit event	Transcription produced — model, version, confidence
Failure path	Low confidence → flagged for human attention. Failure → the report **continues without a transcript**; the audio is still evidence

Honest expectation setting. Published Urdu ASR results are not directly transferable to this setting. On a read-speech Urdu corpus, one 2023 study reports 13% word error and 31.2% sentence error for an HMM-GMM system and 5.2% word error and 20.45% sentence error for a CNN-TDNN system (arXiv:2307.12759). Those figures come from **clean read speech**, not roadside speech in rain and traffic with code-switching. Real accuracy must be measured on Parakh's own pilot data before any claim is made. This is why the audio itself remains the evidence and the transcript is only an aid.

STEP 009 — Categorisation, severity and duplicate detection

Role	AI system
Action	Suggest a category and severity; find possible duplicates
System response	*Suggested category: road damage / pothole. Suggested severity: high — pedestrian route, near school. Possible duplicates: 2 candidates within 40 m*
AI role	**Suggest only.** Every output carries a confidence value and a model version
Human role	**A human confirms or overrides every one of these.** No AI suggestion changes a state, closes a case, rejects a report, or judges a person
Status before	**PROCESSING**
Status after	**AI_PROCESSED** (T3)
Audit event	Annotations written, each attributed to model and version
Failure path	Model unavailable → the report moves to human review **without annotations**. Degraded AI never blocks a case

> **The rule that governs this entire phase:** *AI accelerates human judgement. It never replaces it.* AI cannot verify, approve, reject, close, prioritise into action, or label a citizen. Those are human decisions with human names attached (DR-16).

Phase C — Human Moderation

STEP 010 — Moderator triage

Role	Moderator
Screen	Moderation queue [M-01]
Action	Triage the report
What the moderator sees	Original media · transcript · AI annotations with confidence · location · duplicate candidates. **Not the reporter's identity**, except through a separately authorised, logged and justified access path
Validation	Is this a real, actionable civic issue? Is it a duplicate? Is there a safeguarding concern?
System response	Accepted → a **structured issue** is created and enters **OPEN_UNVALIDATED**. Rejected → **REJECTED** with a reason. Duplicate → merged, supporter count incremented
AI role	Advisory only
Human role	**The decision.** Named, timed, reasoned
Status before	**AI_PROCESSED**
Status after	**MODERATION_APPROVED** (T4a) or **REJECTED** (T4b)
Notification	Aisha is told her report is accepted (with the issue reference), or rejected (with the reason and the appeal path)

Audit event	Moderation decision — actor, time, reason
Next responsible role	Street Representative
Failure path	Safeguarding concern → **immediate** escalation, 24-hour route, content restricted. Ambiguous → request more information. Backlog → ageing is visible and reported

STEP 011 — The offline case (parallel path)

Situation	Aisha has no signal at the roadside
What happens	Capture, location and voice all work offline. The bundle is stored **encrypted** on the device and queued
Integrity	Hash and metadata are computed **at capture**, not at upload — so an offline report is exactly as trustworthy as an online one
Sync	Uploads automatically when connectivity returns
What Aisha sees	A clear pending-sync state on her case
Timestamp	**Capture** time is what counts, not upload time
Failure path	Device lost before sync → the report is lost. This is a known, accepted limitation of not sending unencrypted data off-device

Phase D — Human Verification

STEP 012 — Assignment to a representative

Role	System → Street Representative
Action	Route the issue to a scoped representative
Validation	Holds the geographic scope · **did not report it** · not named in it · no declared conflict · load-balanced
System response	Assigned, with a 72-hour benchmark
Status	**OPEN_UNVALIDATED**
Notification	The representative is notified
Audit event	Assignment recorded
Failure path	No eligible representative → moderator escalation. No action in 72 hours → returns to the pool, moderator notified

STEP 013 — Field validation

Role	Street Representative
Screen	Validation packet [SR-03]
Action	Confirm the problem is real
Input	A **required note**; optional supplementary live-captured photographs; optional local context and priority input
What the representative sees	Everything about the problem. **Nothing about the reporter**
System response	Outcome recorded. A moderator closes the state
AI role	None in the decision
Human role	Independent observation — the step that converts *reported* into *verified*
Status before	**OPEN_UNVALIDATED**
Status after	**VERIFIED** (T5) or **NOT_VALIDATED** (T6)
Notification	Aisha and supporters are told the issue is verified, or told it was not validated with the appeal path
Audit event	Validation recorded — actor, outcome, note, time
Next responsible role	Action owner
Failure path	Conflict → recusal, reassignment, logged. Site unsafe → `unsafe_to_verify`, which **raises** severity. Outside scope → reroute. Already fixed → noted [handling TO BE CONFIRMED]

STEP 014 — Community corroboration (parallel)

Role	Other citizens
Action	Support the issue
Validation	**One support per verified account.** Supporters are counted, not named publicly
System response	Support count rises; it is an input to prioritisation, never an automatic trigger
Human role	Neighbours confirm they are affected
Audit event	Support recorded
Failure path	Repeat attempt → rejected. Coordinated inauthentic support → detected, flagged, **human-reviewed**

Phase E — Institutional Action FUTURE

> Steps 015 to 018 describe the **future institutional model**. In the **Phase-1 pilot**, the action owner is the **pilot-site authority** — school, NGO or community group — and a moderator or administrator records acknowledgement, work-start and fix-claimed **by proxy**, under audit. The state machine is identical. Only who holds the capability changes.

STEP 015 — Officer review and acceptance

Role	Union Council Officer F
Screen	Verified issue detail [UC-04]
Action	Review the evidence bundle and accept the issue
What the officer sees	Original evidence · transcript · AI annotations with provenance · the representative's note · verified supporter count · duplicate and recurrence history · full audit trail. **Not the reporter's identity**
System response	Accepted into the operations workflow
Status before	**VERIFIED**
Status after	**IN_PROGRESS** on acknowledgement or work start (T7)
Notification	Citizen, supporters and representative
Audit event	Officer decision — actor, time, reasoning
Failure path	Not a municipal responsibility → reroute with a recorded reason. Late acknowledgement → the 7-day benchmark is **reported**, not enforced

STEP 016 — Work order and contractor assignment

Role	Officer / procurement authority F
Action	Record scope, cost, budget reference, timeline and milestones; link the authorised contractor assignment
Validation	Cost referenced against the applicable rate card; outliers and duplicate orders **flagged for human review**
AI role	Anomaly flagging. **A flag is a prompt to look, never an accusation and never a rejection**
Human role	Every decision
Audit event	Work order recorded
Failure path	Anomaly flagged → human review. Wrong unit → reassignment, with the correction preserved in the trail

> **Parakh does not select the contractor.** The procurement authority does that, under its own rules, in its own systems. Parakh records or links the result.

STEP 017 — Contractor completion evidence

Role	Contractor F
Screen	Work order detail → capture [CT-02]
Action	Submit completion evidence
Input	Live photographs from the **same location** and the **same or sufficiently similar angle** as the before evidence; optional voice note; milestone selection

Validation	Live capture · GPS proximity · timestamp within the work window · angle similarity · integrity hash · metadata completeness
System response	Evidence attached to the work order and to the issue
AI role	Before/after comparison — **describes** differences, **flags** mismatches. Advisory
Human role	A human judges acceptability
Status before	**IN_PROGRESS**
Status after	**FIX_CLAIMED** (T8)
Notification	Citizen, supporters and representative are invited to confirm
Audit event	Completion evidence submitted — named person, time, all check results
Failure path	Gallery attempt → blocked. Wrong location → flagged. Outside the window → flagged. Angle mismatch → flagged for human review, not auto-rejected

STEP 018 — Officer decision on a milestone FUTURE / PHASE 3

> Where an institution operates a milestone workflow, an **authorised institutional officer** reviews verified milestone evidence **inside Parakh** and records one of: **APPROVE**, **PARTIALLY APPROVE**, **HOLD**, or **REJECT**, with a mandatory reason.
>
> **Parakh never holds, pools, stores, or independently transfers public money.** The recorded decision is an **instruction to the authority's own financial system**, which executes it entirely outside Parakh. See §10.

Phase F — Confirmation and Closure

STEP 019 — Citizen confirmation

Role	Citizen (Aisha) and supporters
Screen	Case detail → confirm [C-07]
Action	Confirm or dispute the claimed fix
Input	Confirm, or dispute with a reason and optional live-captured photograph
System response	Confirmed → **RESOLVED**. Disputed → back to **IN_PROGRESS**
AI role	None
Human role	**The person affected has the final word on whether their problem was solved**
Status before	**FIX_CLAIMED**
Status after	**RESOLVED** (T9) or **IN_PROGRESS** (T11)

Notification	All parties
Audit event	Confirmation or dispute recorded
Failure path	No response within the confirmation window → [POLICY DECISION REQUIRED — whether silence closes the case or escalates to the representative. This must be decided before pilot; silence is not consent by de

STEP 020 — Closure and archive

Role	System
Action	Close and archive the case
System response	Case closed. The full evidence chain — original media, hashes, annotations, human decisions, notifications — is retained under the retention schedule
Status before	**RESOLVED**
Status after	**CLOSED / ARCHIVED** (T10)
Notification	Final confirmation to Aisha
Audit event	Closure recorded; the hash chain over the case is complete and independently checkable
Failure path	Reopened later within the defect liability period → linked as a defect claim (§8.5)

STEP 021 — What remains afterwards

The complete, tamper-evident evidence chain	Anything editable after the fact
Every human decision with actor, time and reason	Any silent deletion — there is no such capability
Recurrence history at this location	Reporter identity in any operational view
Contractor performance signal F	Personal data beyond the retention schedule
Aggregate pattern data for the private operational heatmap	Any public exposure of individuals

Twenty-one steps, one principle. At every point where something consequential happened — accepted, verified, actioned, claimed fixed, confirmed, closed — **a named human made the decision** and **the record cannot be quietly changed**. That is the whole design.

10. The Commit–Verify–Release Model FUTURE / PHASE 3

> **Nothing in this section is built, deployed, or active.** It describes a **future institutional model** that requires a formal partnership, a legal framework, and a government financial system. It is documented here so that everyone using Parakh understands the boundary **before** anyone is tempted to cross it.

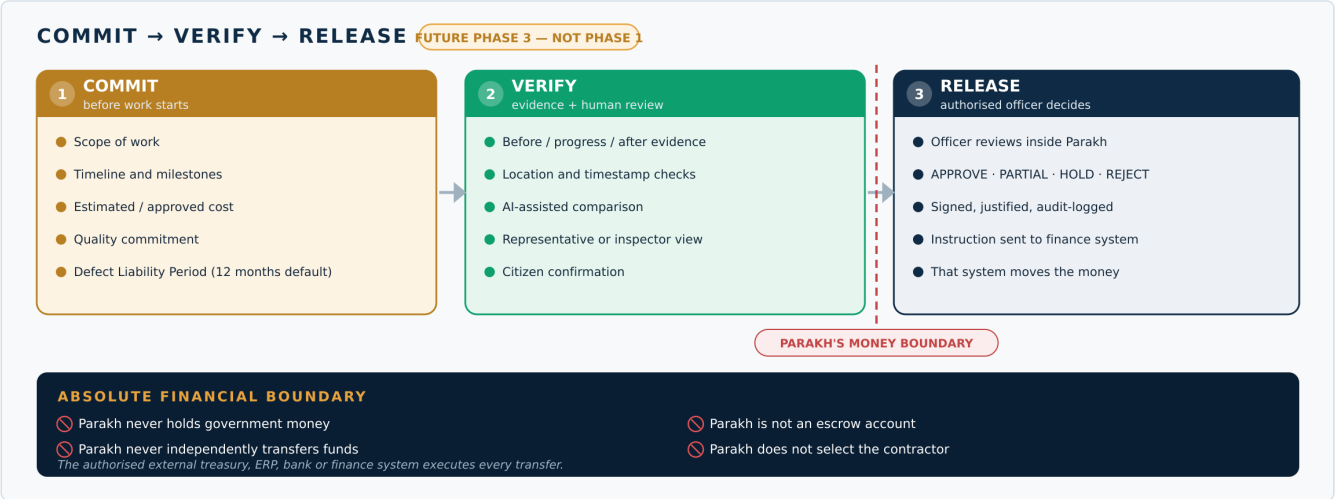


Figure 10.1 — The three stages, the money boundary, and the four absolute prohibitions.

10.1 The problem this model addresses

In a conventional works process, payment often follows a **paper certificate of completion**. The certificate says the work was done. Frequently, nobody independently checked, and when the road fails six months later the trail has gone cold.

The Commit–Verify–Release idea is simple: **make the release of a payment depend on verified evidence rather than on an assertion.**

10.2 The three stages

1 · COMMIT

The authority commits funds against a defined work item, with scope, cost, timeline and milestones recorded in Parakh.

The money stays in the authority's own financial system.

2 · VERIFY

The contractor submits live-captured milestone evidence. Automated checks run. **A human reviews.** A representative or inspector may corroborate. Citizens may confirm.

This is where the evidence is tested.

3 · RELEASE

An **authorised institutional officer** reviews the verified evidence **inside Parakh** and records: **APPROVE · PARTIALLY APPROVE · HOLD · REJECT** — with a mandatory reason.

The authority's financial system executes the decision. Outside Parakh.

10.3 The money boundary — stated as plainly as possible

> **PARAKH NEVER HOLDS, POOLS, STORES, OR INDEPENDENTLY TRANSFERS PUBLIC MONEY.**
>
> Parakh is **not** a bank, an escrow agent, a payment processor, a treasury, or a disbursement system. It has no wallet, no account balance, no transfer capability, and no automated release.
>
> What Parakh does: **records a documented officer decision, grounded in verified evidence, with a complete audit trail.**
>
> What the authority does: **holds the money and executes the decision, in its own system, under its own controls and its own law.**

10.4 The four absolute prohibitions

#	Prohibition
1	**No AI approves a payment.** AI may compare before-and-after evidence and describe what it observes. It may flag anomalies. It never approves, never releases, never certifies.
2	**No automatic release.** No threshold, no score, no timer, no volume of confirmations releases a payment. A named officer decides, every time.
3	**No money inside Parakh.** No custody, no pooling, no transfer, no balance.
4	**No self-approval.** The person who submits evidence can never be the person who approves against it. Separation of duties is structural, not a policy suggestion.

10.5 Officer decision options

Decision	Meaning	Required
APPROVE	Evidence supports full milestone completion	Mandatory reason; audit entry
PARTIALLY APPROVE	Some of the milestone is verifiably complete	Mandatory reason stating what is and is not accepted
HOLD	Evidence is insufficient or a question is outstanding	Mandatory reason; the contractor can respond
REJECT	Evidence does not support the claim	Mandatory reason; appeal route available

Every one of these is recorded with the officer's identity, the time, the evidence reviewed, and the reason — in an append-only trail that cannot be edited afterwards.

10.6 What must be settled before this is ever built

Legal authority for evidence-linked release	An officer cannot lawfully condition a payment on a private platform's record without a legal basis
Integration model with the authority's financial system	Determines whether Parakh's decision record can be relied on at all
Appeal and dispute process for contractors	A decision with no appeal is not a fair decision
Liability allocation	If verified evidence later proves wrong, who bears the loss?
Independent audit of the decision trail	The trail must be checkable by someone other than the platform
Anti-collusion controls	Officer, inspector and contractor relationships must be structurally separated

Until every one of these is resolved with a partner institution and legal review, this model stays exactly where it is: on paper.

11. Portal 5 — Mayor & Operations Command PHASE 1 (LIMITED)

FULL: FUTURE

In Phase 1 this portal exists in a **limited administrative form** used by the pilot's own operations team. The full executive command view — jurisdiction-wide, multi-department, with institutional roll-ups — is a **future capability**.

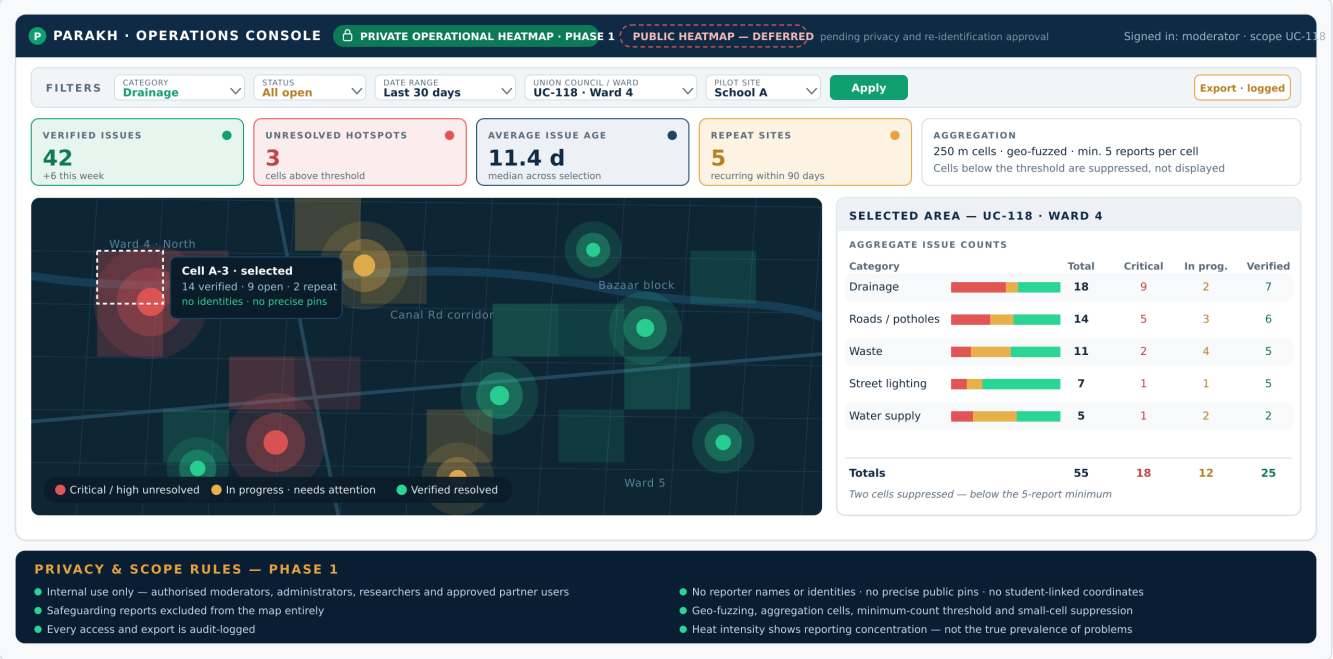


Figure 11.1 — The Phase-1 operational heatmap. Private, internal, and deliberately not a public map.

11.1 What this portal is for

Individual cases are handled in the portals above. This portal answers **different questions**:

- Where are problems **concentrated**?
- What is **ageing** past benchmark, and where?
- Which categories are **rising**?
- Which locations **recur** after being marked fixed?
- Where is the **verification capacity** thin?
- Is the platform itself **behaving properly** — queue depth, review latency, AI drift, anomaly volume?

11.2 The operational heatmap — and why it is private

The heatmap is an internal operational tool. It is not published, and there is no public map in Phase 1.

Why:

- A public map of "problem areas" attaches a **stigma to neighbourhoods** and to the people who live in them.
- Density on a map reflects **who reported**, not where problems actually are. Publishing it would misinform.
- Fine-grained public location data can expose reporters in **small communities**, which breaks the identity protection the whole design rests on.
- A public leaderboard of underperforming areas creates pressure to **suppress reports** rather than fix problems.

Aggregation and geo-fuzzing apply even internally. Points are aggregated into cells and coordinates are deliberately blurred, so that no view is precise enough to identify an individual reporter or a specific household. Sensitive locations — schools, shelters, health facilities and any site flagged by the pilot partner — receive additional protection. [Minimum cell size, blur radius and suppression thresholds TO BE CONFIRMED before pilot]

A public live heatmap is deferred. It is not a Phase-1 capability and must not be built or published without a separate decision, a privacy assessment, and partner agreement.

The visual status model

Colour	Meaning	Typical states
RED	Unresolved or critical — nothing is happening, or the issue is dangerous	VERIFIED but unassigned · overdue · `unsafe_to_verify` · reopened after dispute
YELLOW	Ongoing — the issue is being acted on	IN_PROGRESS · FIX_CLAIMED awaiting confirmation
GREEN	Verified or resolved — a human has confirmed the outcome	RESOLVED · CLOSED / ARCHIVED

Colour is never the whole story. A green cell means a human confirmed a fix; it does not mean the area has no problems, and it does not mean nobody is still waiting. Always open the underlying counts before drawing a conclusion.

11.3 Dashboard components

[SCREEN PLACEHOLDER OP-01 — Operations Command]

This screen must let the user: see the operational heatmap for the jurisdiction; see counts by state; see ageing against benchmarks; see category distribution and trend; see recurrence hotspots; see review capacity and latency; see anomaly flags; and drill from any aggregate into the underlying cases they are authorised to view.

Component	What it shows	What it must never show
Heatmap	Coarsened density of verified issues	Individual reporter locations
State counts	How many cases sit in each state	—
Ageing	Time in state against the 72-hour / 7-day benchmarks	—
Category trend	Which categories are rising or falling	—

Component	What it shows	What it must never show
Recurrence	Locations where a "fixed" problem returned	—
Capacity	Representative coverage, median validation time, agreement rate	Published individual scorecards
AI health	Confidence distribution, override rate, drift indicators	—
Anomaly flags	Cost outliers, duplicate orders, evidence-check failures	Any automated accusation

Filters and roll-ups

Filter	Use
Category	Roads · drainage and sewerage · garbage · streetlights · other supported categories
Geography	Ward → Union Council → jurisdiction roll-up F
Department / operational unit F	Which unit owns the backlog
Project / work order F	Scope · budget reference · milestones · delays · DLP and rework history
Contractor F	Linked project information, **only where the viewer is authorised** and subject to the visibility decision in §8.12
State and age	Everything overdue against the 72-hour and 7-day benchmarks

UC performance roll-ups F compare Union Councils on verification latency, acknowledgement latency, recurrence rate and dispute rate. They are **operational diagnostics, not league tables**, and they are not published.

Four layers that must never be confused

1 · DATA

What was captured and recorded: evidence, locations, times, states, decisions. Factual and auditable.

2 · AI-ASSISTED INSIGHT

Transcripts, suggested categories, anomaly flags, clustering. Probabilistic, versioned, and **always labelled as such**.

3 · HUMAN DECISION

A named person judges, with a reason, inside Parakh. This is where accountability attaches.

4 · OFFICIAL ACTION

What the institution actually does — assign, repair, pay, enforce — in its own systems, under its own authority.

There is no overall AI "health score". Parakh does not produce a single composite number that grades a ward, a Union Council, an officer or a contractor, and senior officials are not asked to rely on one. Composite scores hide their own assumptions, cannot be argued with, and invite decisions nobody can explain afterwards. Every indicator here is a **specific, inspectable count** that can be drilled into.

11.4 Reading the metrics honestly

Three traps, and how to avoid them.

1 · More reports does not mean more problems. It usually means more **trust** or better coverage. A ward whose report count triples after a school partnership has not decayed; it has started speaking.

2 · A quiet area is not a healthy area. Silence can mean no problems — or no coverage, no connectivity, no trust, or fear. Treat low volume as a **question**, not an achievement.

3 · Do not manage the number. The moment a case count becomes a target, the incentive shifts from fixing problems to suppressing reports. Every metric here exists to prompt investigation, never to grade a community.

11.5 Escalation and intervention

Signal	Appropriate response
Verification queue ageing past benchmark	Recruit or reassign representatives in that scope
A category rising sharply	Investigate the cause before assuming a reporting artefact
Recurrence at the same location	Treat as a **quality** problem, not a reporting problem
Representative agreement rate drifting	Calibration and retraining — not punishment
Anomaly flags clustering on one entity	Human review through the proper channel
Safeguarding escalations rising	Immediate review with the pilot partner

11.6 What this portal cannot do

CANNOT

- Publish the heatmap
- See reporter identity
- Edit or delete evidence
- Alter the audit trail
- Override a citizen's confirmation
- Approve payments
- Assign contractors

CAN

- See aggregate patterns for its jurisdiction
- See platform-health indicators
- Drill into authorised cases
- Direct operational capacity
- Escalate to the pilot partner
- Export aggregate reports [format TBC]

MUST

- Treat every metric as a question
- Respect aggregation thresholds
- Investigate quiet areas, not just loud ones
- Record the reason for any intervention

AUDIT

Every drill-down into case-level data is logged with actor, time and purpose. Access to the identity vault is separately authorised, separately logged, and requires a stated justification — for **every** role, including this one.

12. Quick-Start Guides

One page per role. These are the pages to print and hand to someone on their first day.

12.1 Citizen — Quick Start

In one sentence: *Photograph the problem, say what is wrong, and Parakh turns it into evidence someone has to answer.*

Before you start: a verified account (§5.3). Anonymous reporting is not possible.

1	Open Parakh and start a report
2	**Photograph the problem while standing in front of it.** The camera opens directly — there is no gallery upload
3	Confirm the location on the map
4	**Speak** your description. **Urdu, English, or a mix of the two** — all fine at launch
5	Submit. You receive a case reference
6	Follow your case as it moves through verification and action
7	When a fix is claimed, **confirm or dispute it.** Your confirmation is what closes the case

Remember

- Your identity is verified but **never shown** to reviewers, officers, contractors or the public.
- No signal? Capture anyway — it syncs later, and the **capture time** is what counts.
- Only one support per person per issue.
- Deliberately false reports have consequences, decided by a human, with an appeal.

Under 18? Report only through a school or supervised programme, with consent, and with an adult in the loop. Never go somewhere unsafe to take a photo.

Escalation path. Report not accepted → use the appeal route shown in your case. Safety concern → tell the moderator through the case, or contact emergency services directly for anything urgent. Something wrong with your account → platform administrator.

Security reminder. Sign out on any shared device. Your account is tied to your verified identity, and everything submitted under it carries your name in the vault.

12.2 Street Representative — Quick Start

In one sentence: *You confirm, independently, that a reported problem is real — and that is what makes it verified.*

1	Open your validation queue for your scope
2	Open an item. Read the evidence, transcript and AI notes
3	**Go and look**, where practical
4	Record **VALIDATED** or **NOT_VALIDATED** — with a **required note saying what you actually checked**
5	Add local context: how long, who is affected, whether it recurs, whether it is dangerous
6	Attach supplementary live-captured photos if they help
7	Act within **72 hours**, or the item returns to the pool

Remember

- You **never** see who reported anything. Do not try to work it out.
- Recuse yourself from anything involving you, your family, or a declared interest. Recusal is good practice.
- You **recommend**; a moderator closes the state.
- Site unsafe? Mark `unsafe_to_verify` — that raises severity rather than hiding the problem.
- You cannot delete or hide a report. That capability does not exist.

Escalation path. Safeguarding → moderator, immediately. Conflict → recuse. Outside scope → reroute suggestion. Disagreement with a moderator → record your dissent. Pressure from anyone to change a validation → platform administrator.

Security reminder. Sign out after every session. Your grant is scoped and time-bounded, and every action carries your name.

12.3 Union Council Officer — Quick Start FUTURE

In one sentence: *You receive problems that have already been verified by two independent humans, and you decide what the institution does about them.*

1	Sign in with multi-factor authentication
2	Review the verified issue queue for your jurisdiction
3	Open the evidence bundle: media, transcript, AI notes, validation note, supporter count, recurrence, audit trail
4	Accept the issue into the operations workflow, or reroute it with a recorded reason
5	Assign the responsible unit or officer
6	Record scope, cost, budget reference, timeline and milestones
7	Link the authorised contractor assignment made by the procurement authority

Remember

- Only **verified** issues reach you. That is what the two review layers are for.
- You do **not** see reporter identity.
- **Parakh does not select contractors** and does not approve contracts.
- Anomaly flags are prompts to look, never accusations.
- A wrong assignment should be corrected — the correction stays in the trail, and that is the point.

Escalation path. Not a municipal responsibility → reroute with a recorded reason. Above your delegation → the next authority level under your institution's escalation policy. Anomaly you cannot resolve → human review through the proper channel.

Security reminder. Multi-factor authentication is mandatory. Sign out fully; never leave an authenticated session open on a shared terminal.

12.4 Contractor — Quick Start FUTURE

In one sentence: *Do the work, then prove it from the same spot the problem was photographed from.*

1	Sign in and open your assigned work orders
2	Read the scope, timeline, milestones and quality commitment
3	**Study the before evidence** — you will have to match it
4	Do the work to the specified standard
5	**At the site**, open the work order and capture live completion evidence
6	Match the original **location** and **viewing angle** as closely as the reference allows
7	Submit per milestone. Add a voice note describing what was done

Remember

- **No gallery uploads.** There is no such path.
- Checks run on location, time, angle and integrity. A failed check means **human review**, not automatic rejection.
- **FIX_CLAIMED is not RESOLVED.** Only human confirmation closes a case.
- A **defect liability period applies — default 12 months**, as stated in your work order: failures caused by poor execution are corrected at your own cost.
- You cannot confirm your own work, and you cannot edit submitted evidence.
- **No money moves through Parakh, ever.**

Escalation path. Scope query or site condition differs → variation request to the assigning officer. Rejected submission you believe is wrong → attach a documented response and escalate to the officer. Defect claim you dispute → documented response to the authority.

Security reminder. Sign out on site devices. Every submission is attributed to a named person, not to the company.

12.5 Mayor / Operations — Quick Start

In one sentence: *You are not managing cases; you are reading patterns and directing capacity.*

1	Open the operations dashboard for your jurisdiction
2	Check ageing against the 72-hour and 7-day benchmarks first
3	Read the heatmap for concentration — and remember it reflects reporting, not reality
4	Check recurrence hotspots: a "fixed" problem that returned is a quality failure
5	Check verification capacity and latency; thin coverage is an operational risk
6	Review platform health: queue depth, AI confidence, override rate, anomaly volume
7	Act on capacity, escalate what needs the pilot partner, and record your reason

Remember

- **The heatmap is private.** It is never published.
- More reports usually means more trust, not more decay.
- A quiet area is a **question**, not a success.
- Never turn a case count into a target — that incentivises suppression.
- Every drill-down into case data is logged, including yours.

Escalation path. Capacity gap → recruit or reassign. Systemic delay → the pilot partner. Safeguarding volume rising → immediate review with the partner. Suspected integrity failure → platform administrator and security incident process.

Security reminder. Every drill-down into case-level data is logged, including yours. Sign out fully after each session.

13. Training Scenarios and Role Readiness

Reading this manual is not the same as being able to operate a portal. This section is the practical part: scenarios to work through, and checklists that decide whether someone is ready.

13.1 How to run a training session

Group size	6–12. Small enough that everyone speaks
Duration	Citizen: 45 minutes · Representative: half a day · Moderator: full day
Method	Read the scenario aloud → each person writes their answer alone → discuss → compare against the model answer → record disagreements
The disagreements matter most	Where trainees disagree is exactly where the platform will produce inconsistent decisions in the field
Materials	This manual · the relevant quick-start card (§12) · the exception index (§14)
Pass condition	The role-readiness checklist below, completed and signed by a trainer

13.2 Citizen scenarios

S-C1 · The gallery photo. Aisha photographed the pothole yesterday on her phone's normal camera. Today she opens Parakh and looks for a way to upload it.

What should happen? There is no upload path. She must return to the site and capture live. **Why:** a gallery image proves nothing about where or when it was taken, and the whole platform rests on that proof. **Trainer note:** trainees often find this frustrating. Sit with the frustration — it is the correct response, and understanding *why* prevents workarounds later.

S-C2 · No signal. Bilal is on a rural road with no data connection.

What should happen? He captures and describes normally. The bundle is encrypted on his device and queued. It uploads when he reconnects. **The capture time is what counts, not the upload time. Trap:** trainees often assume they must wait for signal before capturing. They must not — capture first.

S-C3 · Someone already reported it. Fatima reports a broken streetlight; the system finds a near-identical open issue 20 metres away.

What should happen? A human moderator decides whether it is a duplicate. If merged, Fatima becomes a **supporter** and her count is added. Her report is not lost, and she is not accused of anything. **AI suggested the duplicate; AI did not merge it.**

S-C4 · The 15-year-old. A student notices a collapsed drain cover on her walk home and wants to report it.

What should happen? Only through her school's supervised programme, with consent, with an adult in the loop, and never by going somewhere unsafe. **Students are never Street Representatives** and are never asked to validate anyone else's report. If in doubt, she tells a teacher and the teacher decides.

S-C5 · The fix that is not a fix. A contractor claims the pothole is fixed. Aisha walks past and sees loose gravel tipped into the hole.

What should happen? She **disputes** the claimed fix, with a reason and a live photograph. The issue returns to **IN_PROGRESS**. **Her confirmation is the gate** — a claim by the party who did the work never closes a case on its own.

S-C6 · The angry report. A citizen submits a report whose voice note is mostly an accusation against a named local official.

What should happen? The moderator assesses the **civic issue**, not the accusation. Parakh is not a complaint channel about individuals (§2.5). If there is an actionable infrastructure problem, it proceeds; if the submission is only an allegation about a person, it is rejected with a reason and an appeal path. **No AI labels anyone.**

13.3 Street Representative scenarios

S-R1 · You think you know who reported it. Reviewing a packet about a drain outside a particular house, you are fairly sure who submitted it.

What should happen? If that belief affects your judgement, **recuse**. If it does not, proceed and judge only the problem. Never record, repeat or act on the guess. The packet withholds identity by construction — your inference is not a licence to reintroduce it.

S-R2 · The contractor is your cousin. A validation request concerns work an entity related to you would likely bid for.

What should happen? **Recuse.** The item returns to the pool and is reassigned. The recusal is logged — and it counts in your favour, not against you.

S-R3 · The problem is already gone. You visit the site and the pothole has been filled by someone.

What should happen? Record what you found, with a note.

[WORKFLOW DETAIL TO BE CONFIRMED — whether this closes the issue or routes to a moderator. Until confirmed, route it to a moderator and say exactly what you saw.]

S-R4 · You cannot safely get there. The reported site is on a flooded road after heavy rain.

What should happen? Mark `unsafe_to_verify`. This **raises** the issue's severity for review. It does not make the issue disappear, and it does not count against you. **Never** put yourself at risk to complete a validation.

S-R5 · The AI is confident and wrong. The AI suggests *"road damage — high severity"* with high confidence. On site you find a shallow surface crack.

What should happen? Record what you observed. High confidence is not evidence — the model has never been to the street. Your override is exactly the value you add, and it feeds calibration.

S-R6 · A welfare concern in the evidence. The voice note mentions a child being hurt.

What should happen? **Stop and escalate immediately** through the safeguarding route — 24-hour escalation applies. Do not continue the ordinary validation as if nothing happened. Safeguarding outranks workflow, always.

13.4 Moderator scenarios

S-M1 · Pressure to remove a report. Someone locally influential asks you to make a report go away.

What should happen? You cannot. There is **no silent deletion capability**. Any rejection you record carries your name, the time, and your reason, and is visible in the audit trail and appealable. Say so plainly, and report the approach to the platform administrator.

S-M2 · Coordinated support. Twenty support actions arrive on one issue within four minutes.

What should happen? The pattern is flagged. **A human reviews it.** Coordinated inauthentic support is not automatically punished, and genuine community concern is not automatically suspicious. Investigate; record what you find.

S-M3 · Identity vault access. A police request arrives concerning a reporter.

What should happen?

[POLICY DECISION REQUIRED — the lawful-request procedure, approval authority and notification rules must be established with legal counsel before pilot.]

Until that exists: **do not disclose**, escalate to the platform administrator and the pilot partner, and log the request. Any vault access is separately authorised, separately logged, and requires a stated justification.

13.5 Role-readiness checklists

A trainee is ready when they can do every item **without looking it up**.

Citizen — ready when they can:

- [ ] Explain why gallery uploads are impossible
- [ ] Capture, locate and voice-describe a report unaided
- [ ] Say what happens to their report offline
- [ ] State that their identity is verified but never shown
- [ ] Find their case and read its status
- [ ] Confirm or dispute a claimed fix, and explain why that matters
- [ ] State the consequence of a deliberately false report, and the right of appeal

Street Representative — ready when they can:

- [ ] State that they never see reporter identity, and why
- [ ] List the four circumstances requiring recusal
- [ ] Write a validation note that says what was actually checked
- [ ] Distinguish original evidence from AI annotations on the packet
- [ ] Explain that they recommend and a moderator closes the state
- [ ] Use `unsafe_to_verify` correctly
- [ ] Route a safeguarding concern within the 24-hour rule
- [ ] Explain the 72-hour benchmark and what happens if they miss it

Moderator — ready when they can:

- [ ] Run a full triage: accept, reject with reason, merge duplicate
- [ ] Explain why no deletion capability exists
- [ ] Identify a safeguarding case and escalate it immediately
- [ ] Explain identity-vault access controls and refuse an improper request
- [ ] Explain that no AI output may reject, restrict or label a citizen
- [ ] Close a validation state and record a proxy action under audit
- [ ] Handle an appeal correctly

Union Council Officer FUTURE **— ready when they can:**

- [ ] Read a full evidence bundle and state what is missing from it
- [ ] Explain that Parakh does not select contractors
- [ ] Record a work order with scope, cost, budget reference and milestones
- [ ] Treat an anomaly flag as a prompt, not an accusation
- [ ] Reassign a wrongly routed issue and explain why the correction stays visible

Contractor FUTURE **— ready when they can:**

- [ ] Capture completion evidence matching the original location and angle
- [ ] Explain why FIX_CLAIMED is not RESOLVED
- [ ] State the defect liability obligation and where its actual length is defined
- [ ] Explain that no money moves through Parakh
- [ ] Respond properly to a rejection or defect claim

Mayor / Operations — ready when they can:

- [ ] Explain why the heatmap is private
- [ ] Explain why rising reports usually indicate rising trust
- [ ] Explain why a quiet area is a question, not a success
- [ ] Read ageing, recurrence and capacity metrics correctly
- [ ] State why case counts must never become targets

14. Exception and Failure Training

Everything below is a situation that **will** occur. Knowing the handling in advance is the difference between a platform people trust and one they route around.

14.1 Capture and evidence exceptions

#	Exception	Handling
E-01	**Gallery upload attempted**	Blocked. No exception, no override, no workaround path exists
E-02	**Camera permission denied**	The app explains why live capture is required and offers to open settings. No report can be submitted without live media
E-03	**Device attestation fails**	The report is **flagged for human review**, not deleted and not silently accepted
E-04	**GPS unavailable or weak**	Manual map placement; report marked `location_unverified` ; **submission is still accepted**
E-05	**Device clock is wrong**	Marked `time_unverified` ; flagged for human review
E-06	**Media integrity hash mismatch**	Treated as tampering evidence. Flagged, retained, human-reviewed. Never quietly dropped
E-07	**Microphone unavailable**	Typed description offered as fallback. Voice is primary, not exclusive
E-08	**Very large media on a slow connection**	Queued and uploaded in the background; the citizen is not blocked
E-09	**Photograph is unclear, dark or out of focus**	Accepted, then flagged at human review. The moderator or representative requests a clearer image rather than rejecting the person
E-10	**Evidence is insufficient to judge the problem**	The reviewer uses the request-more-information path with a **specific** ask. "More information" is not an acceptable request

14.2 Connectivity and device exceptions

#	Exception	Handling
E-11	**No connectivity at capture**	Full offline capture; encrypted on-device queue; syncs later; **capture time governs**
E-12	**Device lost before sync**	The report is lost. A known, accepted limitation of not transmitting unencrypted data
E-13	**App closed mid-report / submission interrupted**	Draft preserved. **DRAFT** is not a submitted state and nobody else sees it. Resume and submit
E-14	**Duplicate submission from a retry**	Detected and merged after human review
E-15	**Duplicate location but materially different evidence**	**Not** merged automatically. A human decides whether it is the same problem or a second, distinct one at the same place
E-16	**Platform or worker outage**	Capture and offline queueing continue on the device. Processing resumes when service returns; nothing queued is lost

14.3 AI exceptions

#	Exception	Handling
E-17	**Transcription confidence is low**	Flagged for human attention. The **audio remains the evidence**
E-18	**Transcription is wrong**	The citizen corrects it before submission; afterwards a reviewer corrects it. Corrections are recorded — the original transcript is not erased
E-19	**Transcription fails entirely**	The report proceeds **without a transcript**. Degraded AI never blocks a case
E-20	**Wrong category suggested**	A human overrides it. The override is recorded and feeds calibration
E-21	**Wrong severity suggested**	Same handling. Severity is a **human** judgement; the representative's local context usually outranks the model
E-22	**AI service unavailable**	Reports route to human review with no annotations. Throughput drops; correctness does not
E-23	**Model drift detected**	Surfaced in platform-health metrics; triggers review, not automatic retraining in production
E-24	**AI produces an inappropriate description**	Human review; annotation corrected; incident logged. The **original evidence is untouched** because AI never modifies it

14.4 Human review exceptions

#	Exception	Handling
E-25	**No eligible representative in scope**	Escalates to a moderator; coverage gap surfaces in operations metrics
E-26	**Validation not actioned within 72 hours**	Assignment expires to the pool; moderator notified
E-27	**Representative has a conflict of interest**	Recusal; reassignment; **logged**. Declared conflicts are checked at assignment, so a conflicted item should never arrive
E-28	**Two representatives assess the same issue differently**	Both assessments are retained. A moderator decides and records why; the dissent stays visible and feeds calibration
E-29	**Representative and moderator disagree**	Dissent is recorded and visible to administrators
E-30	**Reviewer suspects a malicious report**	Flag to moderator; **a human decides**; the citizen may appeal any restriction
E-31	**Citizen appeals a rejection**	Re-reviewed by someone **not involved** in the original decision
E-32	**Safeguarding concern anywhere in the flow**	**Immediate** escalation, 24-hour route, content restricted. Overrides all normal workflow
E-33	**Pressure to suppress a report**	Impossible to comply with — no deletion capability. Escalate to the platform administrator

14.5 Workflow and outcome exceptions

#	Exception	Handling
E-34	**Work assigned to the wrong operational unit**	Reassigned; the original assignment stays visible in the audit trail
E-35	**Officer does not acknowledge within benchmark**	The 7-day benchmark is **reported as a metric**, surfaced in ageing views. Not enforced in Phase 1
E-36	**Contractor delay** F	Recorded against the milestone **with a written explanation**; surfaced to the officer and in ageing views. Penalties belong to the authority
E-37	**After image from the wrong location**	GPS check fails → flagged → human review → resubmission from the correct location
E-38	**After image from a different camera angle**	Similarity check flags it → human review → resubmission where the original angle is reproducible
E-39	**Repair incomplete in the after evidence**	Reviewer records **incomplete**; issue stays **IN_PROGRESS**; rework instruction issued
E-40	**A different defect is visible in the after image**	Recorded as a new observation; a named human decides whether it becomes a new issue or a variation
E-41	**Rework submitted**	Fresh live evidence required. The failed submission is **never deleted** — it stays with its rejection reason
E-42	**Citizen disputes a claimed fix**	Issue returns to **IN_PROGRESS** (T11). Rework, not argument
E-43	**Citizen never responds to a fix claim**	[POLICY DECISION REQUIRED — silence must not default to consent. Decide before pilot]
E-44	**Defect recurs during the DLP**	Linked as a defect claim; **a human** decides whether it is an execution failure, a new problem, or ordinary wear
E-45	**Work stalls indefinitely**	Ageing surfaces in dashboards. **Phase-1 benchmarks are reported, not enforced**
E-46	**Escalation goes unresolved**	Ageing on the escalation itself is surfaced to administrators and the pilot partner [escalation ladder is partner-specific]

14.6 Platform and integrity exceptions

#	Exception	Handling
E-47	**Audit-chain verification fails**	Treated as a **critical security incident**. Off-host signed checkpoints exist precisely to detect this
E-48	**Unauthorised identity-vault access attempt**	Blocked and logged; security incident raised
E-49	**A user's grant expires mid-task**	Access ends at expiry. Grants are time-bounded by design; renewal is a deliberate act
E-50	**System outage / storage or worker tier degraded**	Capture and offline queueing continue on devices; processing backs up and resumes. The **detached worker tier exists so that AI failure cannot take down reporting**. Users see a clear service-status message rather than a silent failure

> **The pattern across all fifty.** Nothing is silently discarded. Nothing is automatically punished. Every ambiguous case ends at a **named human**, and every consequential action leaves a record that cannot be edited afterwards.

15. Privacy, Safety and Trust

15.1 The identity rule

> Identity is **verified** so that reports can be trusted.
> Identity is **never exposed** so that reporters stay safe.

Question	Answer
Who must verify their identity?	**Every** person who submits a report (C-6)
Where is identity stored?	In a **separate encrypted identity vault**, isolated from the operational database (DR-9)
Who sees it in normal operation?	**Nobody.** Not representatives, not moderators, not officers, not contractors, not operations, not the public (DR-18)
How is that enforced?	Identity is removed **in the query that builds each view** — not hidden by a UI setting
Can it ever be accessed?	Only through a separately authorised, separately logged path requiring a stated justification
Under a lawful request?	[POLICY DECISION REQUIRED — procedure, approval authority and user-notification rules must be established with legal counsel before pilot]
Is access reason-based?	Yes. Any access requires a **stated justification** recorded at the moment of access. "Because I have the role" is not a reason
Is access logged?	Yes — **every** protected-data access, for every role including administrators, is written to the append-only audit trail

Identity separation in practice

Layer	Holds	Sees identity
Identity vault	Verified identity, encrypted, isolated	—
Operational database	Reports, issues, states, decisions — linked by pseudonymous participant identifier	No
Evidence store	Write-once media and hashes	No
Reviewer / officer / contractor views	Built by queries that **exclude** identity	No

Layer	Holds	Sees identity
Public	Nothing individual; aggregated and geo-fuzzed only	No

15.2 Safeguarding

Safeguarding overrides workflow. A welfare concern is escalated **immediately** on a 24-hour route, regardless of where the case sits, who raised it, or how busy the queue is.

Minors (C-8): participation is only through supervised institutional programmes, with consent, with an adult in the loop. **Students are never Street Representatives** and never validate reports. No minor is ever asked to enter an unsafe location to collect evidence. Data minimisation applies more strictly to minors than to anyone else.

No CNIC is required from a minor. Where national identity credentials are not lawfully available to or appropriate for a person under 18, the **school-approved identity and consent route** is used instead: the institution attests the participant, guardian consent is recorded, and only the minimum data needed is stored. A minor is never asked for a credential they cannot lawfully hold.

Nobody — of any age — should put themselves at risk to capture evidence. A problem that is dangerous to photograph can be described from a safe distance and marked accordingly.

15.3 Retaliation and pressure

The platform is designed on the assumption that **someone, somewhere, will want a report to disappear**. Its answers:

- **Identity withheld** from everyone who could act on it.
- **No deletion capability** — not for moderators, not for administrators, not for officers.
- **Append-only audit** with off-host signed checkpoints, so tampering is detectable rather than merely prohibited.
- **Every rejection reasoned, named, timed and appealable.**
- **Recusal** as a normal, logged, encouraged act.
- **No public heatmap**, so no neighbourhood is publicly labelled and no reporter is locatable by inference.
- **Aggregation and geo-fuzzing** on every internal map view, with additional protection for sensitive locations.
- **Reason-based, logged access** to any protected data — including by administrators.

15.4 Data handling principles

Principle	In practice
Minimisation	Collect only what an evidence decision requires
Purpose limitation	Evidence is used for civic accountability, not for anything else
Separation	Identity, operational data and media are separately stored
Immutability	Original evidence is write-once; AI annotations are stored separately
Retention	Governed by a published retention schedule [periods TO BE CONFIRMED before pilot]
Portability	S3-compatible storage, so the pilot is not locked to one provider
Aggregation	Internal analytics coarsened so no individual is identifiable

15.5 Ethics review

Independent ethics review is required and is not waivable.

Parakh involves minors, vulnerable communities, identity data and civic accountability in a real jurisdiction. Independent ethics review is a **precondition of pilot deployment**. It is **not** something the founder, owner, or any member of the project may accept, waive, shorten, or self-certify. It is an external check, and the point of an external check is that the project does not get to grade itself.

16. What AI Can and Cannot Do



Figure 16.1 — The line between AI assistance and human decision. It does not move.

AI CAN

- Transcribe voice notes in **Urdu and English, including Urdu-English code-switching** — the approved launch scope (C-4). Punjabi and other regional languages are a planned extension, not a current capability
- Suggest a category
- Suggest a severity
- Propose possible duplicates
- Describe observable differences between before and after evidence F
- Flag cost outliers and duplicate work orders F
- Surface possible recurrence links
- Attach a confidence value and a model version to everything it produces

AI CANNOT

- Verify that a problem is real
- Approve or reject a report
- Close a case or change a state
- Certify that work is complete
- Approve, release, or hold a payment
- Label a citizen fraudulent or malicious
- Restrict, suspend or penalise any user
- Assign a contractor
- See, infer or reveal reporter identity
- Modify original evidence
- Alter the audit trail
- Overrule a human — in any direction

AI accelerates human judgement. It never replaces it.

16.1 How to read an AI output

What you see	What it means	What it does not mean
High confidence	The model's internal certainty is high	That the model is right. It has never been to the site
Low confidence	Look more carefully	That the report is weak or suspicious
A flag	A prompt to investigate	An accusation, a finding, or a rejection
A transcript	An aid to reading the evidence	A replacement for the audio, which is the actual evidence
A duplicate candidate	A suggestion for a human to check	An automatic merge
A model version	Provenance, so a past decision can be audited against the model that informed it	—

16.2 Why the boundary is drawn here

Three reasons, all practical rather than philosophical:

1. **Accountability needs a name.** A decision that affects someone's street, reputation or livelihood must be attributable to a person who can be asked to explain it. A model cannot be asked.
2. **Models fail unevenly.** Speech recognition on code-switched roadside audio in rain is not a solved problem, and its errors will not be distributed fairly across accents and dialects. Placing consequences behind those errors would place them on the people already least well served.
3. **Trust is the product.** A platform whose decisions can be dismissed as *"the computer decided"* has nothing to offer a community that already suspects nobody is listening.

17. Glossary

Every important term, in plain English.

Term	Meaning
Action owner	The party responsible for doing something about a verified issue. Configurable — in Phase 1 the pilot-site authority; in the institutional model, a government unit
After evidence	Live-captured media submitted to show a problem has been fixed, from the same location and a matching angle
AI annotation	Anything a model produces about evidence — transcript, suggested category, suggested severity, duplicate candidate. Stored **separately** from the evidence, with model and version
Appeal	A citizen's or contractor's request that a decision be re-reviewed by someone not involved in making it
Append-only audit	A record to which entries can be added but never edited or removed, hash-chained and checkpointed off-host so tampering is detectable
Before evidence	The original citizen (and representative) media showing the problem, used as the baseline for the after evidence
Capability grant	A scoped, time-bounded permission held by a user. Not a permanent job title
Case reference	The identifier a citizen uses to follow their report
Commit–Verify–Release	The future model linking payment release to verified evidence (§10). Not built
Confidence	A model's own estimate of how certain it is. A signal to look harder — never proof
Defect liability period	Twelve months after completion during which failures caused by poor execution are corrected at the contractor's cost
DLP (Defect Liability Period)	The period after completion during which execution failures are corrected at the contractor's cost. Owner-set default of 12 months, pending legal review and confirmation in the applicable work order or contract
Duplicate	A second report of a problem already recorded. Merged only by a human; the second reporter becomes a supporter
Evidence bundle	The sealed set of media, location, time, hashes and metadata created at submission
FIX_CLAIMED	The state meaning *someone says it is fixed*. **Not** the same as resolved
Geo-fuzzing	Deliberately blurring and aggregating coordinates so a map cannot identify an individual reporter or household
Human-in-the-loop	The design rule that a named person makes every consequential decision, with AI only assisting
Identity vault	The separate encrypted store holding verified identity, isolated from operational data
Issue	The structured, actionable record created when a moderator accepts a report
Live capture	Media taken inside the app, hashed at capture with location and time attached. The only accepted form of evidence
Milestone	A defined stage of a work order that can be evidenced and reviewed on its own

Term	Meaning
Moderator	The human who triages reports, closes validation states, and handles escalations
Original evidence	The media exactly as captured, hashed at capture, never altered and never touched by AI
Proxy action	An action recorded by an authorised moderator or administrator on behalf of a pilot-site authority, under audit, where no institutional actor exists
Recurrence	The same problem reappearing at the same location after closure
Recusal	A reviewer's logged withdrawal from a case they should not judge
Report	What a citizen submits. Becomes an **issue** only after human moderation
Rework	Returning to a site to complete or correct work after evidence was not accepted
Role-based access	Permissions attached to scoped, time-bounded capability grants rather than to permanent job titles
Severity	How serious and urgent an issue is. AI may suggest it; a human decides it
Street Representative	The consented adult who independently confirms that a reported problem is real
Supporter	A verified citizen who confirms an issue affects them. One per person per issue
Union Council	The local-government tier used as the primary jurisdiction unit in the future institutional model
Validator	Any human who independently confirms evidence — in Phase 1, the Street Representative
Verification	Independent human confirmation that a reported problem is real. Distinct from AI processing
Verified issue	An issue independently confirmed by a Street Representative and closed into the VERIFIED state by a moderator
Work order	The authorised record of work to be done: scope, cost, budget reference, timeline, milestones and quality commitment
Write-once storage	Storage in which an object cannot be altered after it is written
`location_unverified`	A flag meaning the location could not be confirmed by GPS. **Does not block a report**
`time_unverified`	A flag meaning the capture timestamp could not be confirmed
`unsafe_to_verify`	A representative's marker that a site cannot be safely reached. **Raises** severity

18. What Is Not Yet Decided

This section exists because a training manual that hides its gaps trains people to be confident about things nobody has decided. Every item here comes from the Master Document's Open Items Register.

18.1 Items that block pilot launch

Area	Open item
Ethics	Independent ethics review — required, **non-waivable**, not yet obtained
Legal	Lawful-request procedure for identity-vault access: process, approval authority, notification
Legal	Confirmation of the responsible tier or department under the **Punjab Local Government Act 2025** — with the partner institution and legal review
Policy	Retention periods for evidence, identity and audit data
Policy	What happens when a citizen does not respond to a claimed fix (E-27)
Policy	Representative probation threshold before unshadowed validation
Privacy	Minimum aggregation cell size and suppression thresholds for the internal heatmap
Safeguarding	Full minors' consent, supervision and data-minimisation protocol, agreed with the pilot partner
Operations	Escalation ladder above the action owner
Product	Handling when a validated problem has already been fixed (S-R3)
Product	Whether contractor performance records are public, authority-only, or tiered

18.2 Everything in this manual marked FUTURE

FUTURE means **designed, not built**. It is documented so that the design is coherent and so that nobody later assumes a capability exists.

- Union Council Officer portal (§7)
- Contractor portal (§8)
- Inspector / Field Verifier role (§7.7)
- Commit–Verify–Release model (§10)
- Full executive Operations Command view (§11)
- Anomaly detection over cost and work orders (§7.8)
- Before/after AI comparison (§8.4)

18.3 The honest status statement

> **Parakh is not an official Government of Punjab platform.** It is an independent civic technology project. Nothing in this manual should be read as a claim of government adoption, endorsement, procurement, or authority.
>
> **Parakh does not currently operate a live pilot.** The design is complete and approved; deployment is not.
>
> **Parakh holds no public money and has no capability to move any.**

19. Manual Completeness Audit

Self-assessment against the required coverage checklist.

#	Required coverage	Where	Status
1	Purpose, scope and audience stated	§1.2–§1.4	Complete
2	Honest status — not a government platform	Cover, §2.6, §18.3	Complete
3	Convention for unconfirmed detail	§1.6	Complete
4	What Parakh is and is not	§2.2, §2.5	Complete
5	Five portals identified and diagrammed	§3.1	Complete
6	Role and permission matrix	§3.4	Complete
7	Master training example used throughout	§3.3, §9	Complete
8	Report and issue states, kept separate	§4	Complete
9	Timing benchmarks stated as benchmarks	§4, §11.5	Complete
10	Citizen portal, step by step	§5.1–§5.18	Complete
11	Live-capture-only rule and its rationale	§5, §8.4, §9 STEP 004	Complete
12	Offline behaviour	§9 STEP 011, E-09	Complete
13	Representative can / cannot / escalate / evidence / privacy / mistakes	§6.13	Complete
14	Reporter identity never visible to reviewers	§6.4, §15.1	Complete
15	UC Officer portal, marked future	§7	Complete
16	Contractor portal, marked future	§8	Complete
17	Before/after same location and angle rule	§8.4	Complete
18	Defect liability period, correctly qualified	§8.9	Complete
19	Full 0–100 case simulation with all required fields	§9, STEP 001–021	Complete
20	Commit–Verify–Release with money boundary	§10	Complete
21	Mayor & Operations portal, private heatmap	§11	Complete
22	Five role quick-start guides	§12.1–§12.5	Complete
23	Training scenarios and readiness checklists	§13	Complete
24	Exception and failure handling	§14, E-01–E-50	Complete
25	Privacy, safety, minors, ethics	§15	Complete
26	AI can / cannot	§16	Complete
27	Glossary	§17	Complete
28	Open items disclosed rather than invented	§18	Complete

#	Required coverage	Where	Status
29	Contractor onboarding, acceptance, progress, delays, variations	§8.2–§8.6	Complete
30	Rework, recurrence and DLP monitoring	§8.8, §8.9, §8.10	Complete
31	RED / YELLOW / GREEN status model	§11.2	Complete
32	Geo-fuzzing and sensitive-location protection	§11.2, §15.3, §15.4	Complete
33	No opaque AI health score	§11.3	Complete
34	Data / insight / decision / action separation	§11.3	Complete
35	Reason-based, logged access to protected identity	§15.1	Complete
36	No CNIC requirement for minors	§15.2	Complete
37	Quick-start escalation paths and security reminders	§12.1–§12.5	Complete
38	Launch language scope stated accurately (Urdu + English, code-switching)	§2.4, §5.9, §9 STEP 006/008, §16	Complete
39	Timeframes verified against the Master Document and labelled	§4.3	Complete
40	Restriction and suspension authorities stated	§4.3, §5.17, §6.9	Complete
41	Defect liability qualified as a default pending legal review	§8.9, §17	Complete

19.1 Deliberate omissions

Not included	Why
Final button names and screen layouts	No user interface exists. Inventing them would train people incorrectly
Statutory provisions of the Punjab Local Government Act 2025	Requires legal review; the Act is recent and this manual will not paraphrase law
A specific adult-literacy percentage for Punjab	The figure previously used could not be tied to an exact official Pakistan Bureau of Statistics table and definition, so it was removed rather than repeated
Any claim of Punjabi-language transcription at launch	Punjabi is a future multilingual extension under C-4, not a Phase-1 commitment
Installation, administration and schema documentation	Out of scope (§1.3); belongs with the Master Document
Any claim of government adoption	Untrue

19.2 Maintenance

This manual is a **living document**. It must be revised whenever a Decision Register entry changes, an open item is closed, a user interface is built, or a pilot produces evidence that the design does not survive contact with the street.

Where this manual and the **Parakh Master Document v1.2.1** disagree, **the Master Document governs**.

End of the Parakh Complete User Training & Operations Manual

Prepared for the Parakh project · Owner: Hassan